

Critical source activation and original observed arithmetic actions

Split-Zero research programme

21 September 2026
SZ-20260921-015

The original source activation now has an evaluated transition on the observed inverse-power space. The complete scalar and joint minima, every original period constraint and the physical coordinate $S = k/2 + iy$ remain unchanged. At the exponential activation

$$\alpha_k(\zeta) = e^{-2q \log(q/k) - \zeta q},$$

the four-cutoff inverse-power determinant has coefficient

$$C_{\text{inv}}(\zeta) = 2[\min(\beta_L, \zeta) - \min(\beta_H, \zeta)], \quad \beta_H = \beta_L - C_\partial/2.$$

The proof passes through a finite harmonic metric enclosure valid at every activation, including the original observation minimum. The exact earlier elliptic identity gives the same C_∂ in this formula. The analytic endpoint inputs retain the proof status and scope of the complete earlier monic-profile and zero-law sources.

The original map M^{v+1} , observed on a specified growing inverse-power subspace, has one small singular direction and retains every other singular direction. Its determinant and late heat are evaluated below with the complete source and target metrics. An explicit shear bound supplies the step abbreviated in the received continuation.

The new full-space bound also applies to the original kernel. It proves exponentially small metric change on the weak side of the critical layer, at the exact original boundary activation as well. The inverse-sector quotient retains the kernel determinant through $o(kq)$ uniformly in activation. Exact determinant-line identities subtract the evaluated inverse-sector contribution from the original complementary return, while retaining both remaining factors.

Complete proofs, the received 48-equation argument, original human equation sources and every earlier provider accompany this edition. The initial canonical kernel coefficient and the original complex-current phases are not assigned values by the positive metric or heat estimates. The following proofs give their exact receiving maps and finite bounds.

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The transition and its original quotient maps

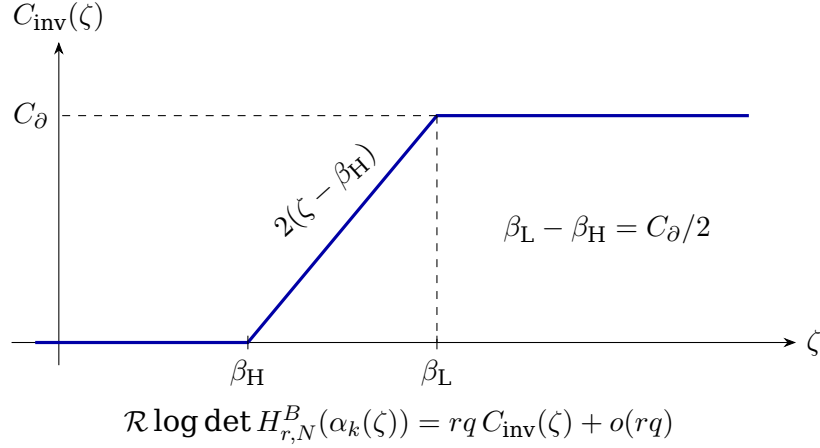


Figure 1: The exact three-branch coefficient, with the original ζ and C_δ labels. Tick spacing is schematic. The companion harmonic proof derives the finite error before taking the limit. AK3 and AK11-12 carry this coefficient into the original determinant factors; no kernel factor is replaced by its rank.

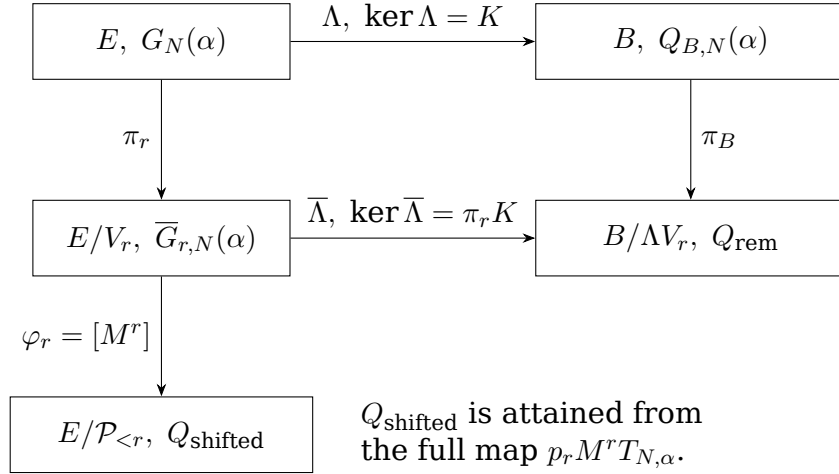


Figure 2: The upper square commutes algebraically. Every quotient metric is the full attained original metric. AK5-6 controls the kernel determinant under the left quotient; AK8-9 proves that φ_r is an isometry for the specified shifted source. AK10 proves the determinant factorization through the lower-right space. The arrow directions state actual domains and codomains.

Part I

Original kernel and complementary receiving proofs

Critical activation carried into the original kernel and complementary quotient

These calculations use the full original source covariance, the unchanged observation and its complete kernel. They strengthen IQR1-6 and the covariance receiving equations without replacing the kernel by the inverse-power space. The all-activation comparison used below is proved in the accompanying harmonic derivation from the original finite IVO estimates; its analytic endpoint coefficient retains the exact earlier monic-profile and zero-law proof status.

1. Original objects and proved input

Keep

$$q = (k+1)^2, \quad k \equiv 1 \pmod{4}, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2, \quad Q(y) = \prod_{a,b=0}^k [y - (2b-k)\gamma + i(2a-k)\delta].$$

Let $E = \mathbb{C}[y]/(Q)$, $M[p] = [yp]$, and retain $S = k/2 + iy$ and the original physical source $w_h^{*k}(y) dy$. Let $\Lambda : E \rightarrow B$ be the original onto observation. Its kernel K , frame I_K , every period constraint and the complete invariant-image columns remain fixed. On the previously proved period domain, $m = \dim K = 8k - 16$.

The original scalar and complete joint source minima are G_N, G_N^J . Their exact activation is

$$G_N(\alpha)^{-1} = (1 - \alpha)G_N^{-1} + \alpha(G_N^J)^{-1}, \quad 0 \leq \alpha \leq 1. \quad (\text{AK1})$$

The joint source contains the scalar source isometrically with the same value map. Taking the infimum over that larger set proves $G_N^J \preceq G_N$. Hence $G_N^J \preceq G_N(\alpha) \preceq G_N$.

Set $U_r d = \sum_{j=1}^r d_j [y^{-j}]$, $V_r = \text{im } U_r$. All finite rational detection guards are retained, and $r \log(q+2) = o(q)$. The proved harmonic comparison is

$$e^{-E_{r,N}} h_N(\alpha) I \preceq H_{r,N}^B(\alpha) \preceq H_{r,N}^E(\alpha) \preceq e^{E_{r,N}} h_N(\alpha) I, \quad h_N(\alpha) = \frac{S_N}{1 - \alpha + \alpha S_N}, \quad (\text{AK2})$$

where $H^E = U_r^* G_N(\alpha) U_r$, $H^B = U_r^* \Lambda^* Q_{B,N}(\alpha) \Lambda U_r$, $Q_{B,N}(\alpha) = (\Lambda G_N(\alpha)^{-1} \Lambda^*)^{-1}$, $S_N = u_k / (|Q(0)|^2 K_{Q,n_0}(0,0))$, and

$$E_{r,N} = O_{h,A}(k \log^2(q+2) + (k+Jr) \log(q+Jr+2)) = o(q).$$

The constants in AK2 are independent of activation. In particular, $H^B \succeq e^{-2E_{r,N}} H^E$. No division of the original arithmetic metric has been made.

Write $\mathcal{R}f_N = f_{q-1} + f_q - f_{2q-1} - f_{2q}$. With

$$\alpha_k(\zeta) = e^{-2q \log(q/k) - \zeta q}, \quad \beta_H = \beta_L - C_\partial/2, \quad C_{\text{inv}}(\zeta) = 2[\min(\beta_L, \zeta) - \min(\beta_H, \zeta)],$$

the harmonic calculation gives

$$\mathcal{R} \log \det H_{r,N}^B(\alpha_k(\zeta)) = r q C_{\text{inv}}(\zeta) + o_{h,A}(r q). \quad (\text{AK3})$$

IQR2 identifies the exact coefficient in this formula with the earlier C_∂ ; this is an algebraic identity of the retained elliptic formulas, not a numerical identification.

2. The original kernel survives the inverse-sector quotient at every activation

Since AK2 makes ΛU_r injective, $K \cap V_r = 0$. The algebraic quotient $\pi_r : E \rightarrow E/V_r$ is fixed. Give it the metric $\overline{G}_{r,N}(\alpha) = (\pi_r G_N(\alpha)^{-1} \pi_r^*)^{-1}$ in any fixed quotient coordinates. Define

$$H_{K,N}(\alpha) = I_K^* G_N(\alpha) I_K, \quad \widehat{H}_{K,r,N}(\alpha) = (\pi_r I_K)^* \overline{G}_{r,N}(\alpha) (\pi_r I_K). \quad (\text{AK4})$$

The image frame has exactly m independent columns because of the intersection statement.

For the following finite calculation fix N, α , and write

$$A = I_K^* G I_K, \quad D = U_r^* G U_r, \quad C = I_K^* G U_r.$$

Minimizing over the full kernel gives $H^B = D - C^* A^{-1} C$. Minimizing over the full inverse sector gives $\widehat{H}_K = A - C D^{-1} C^*$. These formulas follow by expanding $\|I_K x + U_r y\|_G^2$ and completing its quadratic square, so both original minima, rather than a chosen section, are retained.

Put $X = A^{-1/2} C D^{-1/2}$. AK2 implies $I - X^* X \succeq e^{-2E_{r,N}} I$. The nonzero eigenvalues of $X^* X$ and $X X^*$ coincide: if $X^* X v = \lambda v$ with $\lambda > 0$, then $X v \neq 0$ and $X X^*(X v) = \lambda X v$, and the inverse correspondence is X^*/λ . Therefore

$$\begin{aligned} \delta_{r,N}(\alpha) &:= \log \det H_{K,N}(\alpha) - \log \det \widehat{H}_{K,r,N}(\alpha) \\ &= \log \det H_{r,N}^E(\alpha) - \log \det H_{r,N}^B(\alpha) \\ &= -\log \det(I - X X^*) \\ &\in [0, 2\rho_{r,N}(\alpha) E_{r,N}], \quad \rho_{r,N}(\alpha) = \text{rank } C \leq \min(m, r). \end{aligned} \quad (\text{AK5})$$

This also proves the exact relation between the kernel quotient and observed inverse-power metric.

Let $d_N = 2\rho_{r,N} E_{r,N}$. Retaining all four signs yields

$$\begin{aligned} \left| \mathcal{R} \log \det H_{K,N}(\alpha) - \mathcal{R} \log \det \widehat{H}_{K,r,N}(\alpha) \right| &\leq \max(d_{q-1} + d_q, d_{2q-1} + d_{2q}) \\ &\leq 4 \min(m, r) \max_N E_{r,N}. \end{aligned} \quad (\text{AK6})$$

For $r = k$, the right side is $O_{h,A}(k^2 \log^2(q+2)) = o(kq)$, uniformly for every activation, including both endpoints and the critical exponential window. The first inequality uses nonnegativity in AK5; it does not sum absolute errors with an unnecessary factor two. This extends the earlier canonical-source kernel quotient to the complete activated metric.

3. Exact source map after shifting the inverse sector

The quotient used in AK4 has a concrete original source presentation. Let $J_N^{\text{sc}} : \mathcal{X}_N \rightarrow E$ and $J_N^{\text{jt}} : \mathcal{J}_N \rightarrow E$ be the original scalar and joint value maps from their full physical source Hilbert spaces. Define

$$T_{N,\alpha}(p, z) = \sqrt{1-\alpha} J_N^{\text{sc}} p + \sqrt{\alpha} J_N^{\text{jt}} z. \quad (\text{AK7})$$

The direct sum uses the unchanged two physical source norms. Its covariance is exactly AK1, since multiplication by the two real square roots squares their coefficients in $T_{N,\alpha} T_{N,\alpha}^*$. Its entire relation space is $\ker T_{N,\alpha}$. Cross terms arise in the attained minimum and are not removed.

Multiplication by M^r induces the exact isomorphism

$$\varphi_r : E/V_r \longrightarrow E/\mathcal{P}_{<r}, \quad [x]_{V_r} \longmapsto [M^r x]_{\mathcal{P}_{<r}}. \quad (\text{AK8})$$

Indeed M is invertible because none of the original roots is zero, and $M^r V_r = \mathcal{P}_{<r}$; its inverse is multiplication by M^{-r} modulo V_r .

Give the target in AK8 the metric attained from the **same source** by $p_r M^r T_{N,\alpha}$, where $p_r : E \rightarrow E/\mathcal{P}_{<r}$. For any target z ,

$$\begin{aligned} \|z\|_{\text{target}}^2 &= \inf_{p_r M^r T_{N,\alpha} s = z} \|s\|^2 \\ &= \inf_{\varphi_r \pi_r T_{N,\alpha} s = z} \|s\|^2 = \|\varphi_r^{-1} z\|_{\overline{G}_{r,N}(\alpha)}^2. \end{aligned} \quad (\text{AK9})$$

Thus AK8 is an isometry for these explicitly specified attained metrics, and the target Gram of $p_r M^r I_K$ is exactly $\hat{H}_{K,r,N}(\alpha)$. The cutoff remains N . The map does not substitute the unrestricted scalar source at degree $N+r$, whose metric requires the separate IQR5 comparison.

4. The known inverse-sector contribution passes through the determinant lines

Choose a fixed matrix W completing $[I_K, U_r]$ to a basis $\mathcal{T} = [I_K, U_r, W]$ of E . It may depend on k and the original period but is independent of N, α . Write

$$B_r = \text{im}(\Lambda U_r), \quad B_{\text{rem}} = B/B_r,$$

and let $H_{\text{rem},r,N}(\alpha)$ be the metric on B_{rem} attained from the original $Q_{B,N}(\alpha)$, in the basis supplied by ΛW modulo B_r .

Two complete Schur eliminations, first K and then B_r , give

$$\det(\mathcal{T}^* G_N(\alpha) \mathcal{T}) = \det H_{K,N}(\alpha) \det H_{r,N}^B(\alpha) \det H_{\text{rem},r,N}(\alpha). \quad (\text{AK10})$$

Every factor uses its declared original restriction or attained quotient. In particular the final factor contains all corrections along both K and V_r .

The frame factor $|\det \mathcal{T}|^2$ cancels under $\mathcal{R}$, whose four signs sum to zero. Inserting AK3 proves

$$\begin{aligned} \mathcal{R} \log \det G_N(\alpha_k(\zeta)) &= \mathcal{R} \log \det H_{K,N}(\alpha_k(\zeta)) \\ &\quad + \mathcal{R} \log \det H_{\text{rem},r,N}(\alpha_k(\zeta)) + r q C_{\text{inv}}(\zeta) + o(r q). \end{aligned} \quad (\text{AK11})$$

At $r = k$ this subtracts an evaluated $k q$ contribution from the original complementary return.

For any one of the three metric factors, write $\Delta_X(\alpha) = \log \det H_X(0) - \log \det H_X(\alpha)$. For E , this is the full source cost

$$\Delta_E(\alpha) = \log \det(I + \alpha K_N), \quad K_N = G_N^{1/2} [(G_N^J)^{-1} - G_N^{-1}] G_N^{1/2} \succeq 0.$$

Taking the ratio of AK10 at 0 and α proves the exact identity $\Delta_E = \Delta_K + \Delta_{V_r}^B + \Delta_{\text{rem}}$. Consequently

$$\mathcal{R}(\Delta_K + \Delta_{\text{rem}})(\alpha_k(\zeta)) = \mathcal{R} \Delta_E(\alpha_k(\zeta)) - r q [C_\partial - C_{\text{inv}}(\zeta)] + o(r q). \quad (\text{AK12})$$

This is the backward receiver for the earlier full source-metric cost. It does not allocate the unevaluated sum between its two remaining factors.

5. Weak critical activation preserves every original subquotient

The lattice cardinal estimate proves, in the actual original root-value coordinates,

$$\mathrm{Tr}(T_k^{-1}G_N) \leq \Gamma_N S_N, \quad (G_N^J)^{-1} \preceq \mathbf{L}_k T_k^{-1}.$$

For any positive matrix G , set $A = G^{1/2}T_k^{-1}G^{1/2} \succeq 0$. Every eigenvalue of A is at most $\mathrm{Tr} A$, so conjugating this inequality gives $T_k^{-1} \preceq \mathrm{Tr}(T_k^{-1}G)G^{-1}$. Define $c_N = \max(1, \mathbf{L}_k \Gamma_N S_N)$. Thus

$$G_N^{-1} \preceq G_N(\alpha)^{-1} \preceq (1 - \alpha + \alpha c_N)G_N^{-1}, \quad \frac{G_N}{1 - \alpha + \alpha c_N} \preceq G_N(\alpha) \preceq G_N. \quad (\text{AK13})$$

This is a full-space inequality before any projection.

Both restriction and attained quotient preserve a two-sided metric comparison with the same constants: restriction follows by inserting the same vector, while the quotient follows by taking the infimum over exactly the same fibre. Iterating proves the assertion for every fixed original subquotient. For a resulting rank- t metric H_X ,

$$0 \leq \log \frac{\det H_{X,N}(0)}{\det H_{X,N}(\alpha)} \leq t \log(1 - \alpha + \alpha c_N). \quad (\text{AK14})$$

The rank multiplier is retained.

Now $\log c_N = 2q \log(q/k) + \beta_{L/H}q + o(q)$; the equality follows from the proved endpoint expansions and $\log(\mathbf{L}_k \Gamma_N) = o(q)$, with the lower endpoint eventually positive. For $\alpha = \alpha_k(\zeta)$,

$$0 \leq \Delta_{X,N}(\alpha_k(\zeta)) \leq tq[\beta_{L/H} - \zeta]_+ + t o(q). \quad (\text{AK15})$$

The finite controlling quantity is always AK14. For any fixed $\varepsilon > 0$, if $\zeta \geq \beta_L + \varepsilon$, then for all sufficiently large k

$$0 \leq \Delta_{X,N}(\alpha_k(\zeta)) \leq te^{-\varepsilon q/2} \quad \text{at each of the four cutoffs,} \quad |\mathcal{R}\Delta_{X,N}| \leq 2te^{-\varepsilon q/2}. \quad (\text{AK16})$$

Indeed $\alpha c_N \leq e^{-\varepsilon q/2}$, $\log(1 - \alpha + \alpha c_N) \leq \alpha c_N$, and there are two positive and two negative losses, each nonnegative.

In particular this applies to the full original kernel, with $t = m$, and every complementary attained quotient. Since $\beta_L < 2 \log(4/(\pi\gamma)) < 0$, the original boundary activation $\alpha = e^{-2q \log(q/k)}$ has $\zeta = 0$ and satisfies AK16 with $\varepsilon = -\beta_L$. Therefore the canonical kernel determinant at this activation agrees with its unactivated value up to an exponentially small finite metric error. This conclusion uses the full-space cardinal comparison; it does not insert a moving parameter into the earlier fixed-activation asymptotic.

Within $\beta_H < \zeta < \beta_L$, AK14–15 instead give the actual kernel-loss enclosure

$$-2mq(\beta_L - \zeta) - o(mq) \leq \mathcal{R} \log \det H_{K,N}(\alpha_k(\zeta)) - \mathcal{R} \log \det H_{K,N}(0) \leq o(mq). \quad (\text{AK17})$$

The two high-cutoff losses are $o(mq)$, the two low-cutoff losses are nonnegative and at most $mq(\beta_L - \zeta) + o(mq)$ each. This explains both signs in AK17. The exact initial canonical determinant and its individual complex-current coefficients remain separate calculations; AK5–17 give the proved maps and bounds connecting this critical-layer result to them.

6. An exact certificate for the concrete activation coefficient

For the received choice $\zeta_\diamond = \beta_L - a_0$, where $a_0 = \log(4/\pi)$, the coefficient is $C_\partial - 2a_0$. The exact rational calculation accompanying this proof certifies

$$0.870764 < C_\partial - 2\log(4/\pi) < 0.871623, \quad C_\partial > 4\log(4/\pi) > 0. \quad (\text{AK18})$$

Here is its full elementary method. Let $a = \arctan(1/5)$, $b = \arctan(1/239)$. The double-angle formula gives $\tan(2a) = 5/12$, $\tan(4a) = 120/119$ and $\tan(4a - b) = 1$. Both $4a - b$ and $\pi/4$ lie in $(0, \pi/2)$, so $\pi = 16a - 4b$. Integrating the finite geometric series for $1/(1 + x^2)$ bounds each arctangent between its alternating truncation and that truncation plus the next term. The certificate takes 14 terms at $1/5$ and four at $1/239$, giving exact rational endpoints $p_- < \pi < p_+$.

Set $t_- = (4 - p_+)/(4 + p_+)$ and $t_+ = (4 - p_-)/(4 + p_-)$. Integrating $2/(1 - t^2)$ gives $\log((1+t)/(1-t)) = 2 \sum_{j \geq 0} t^{2j+1}/(2j+1)$. Its first 16 terms at t_- are a lower bound for a_0 . The first 16 terms at t_+ , plus $2t_+^{33}/[33(1-t_+^2)]$, are an upper bound. All quantities in this finite calculation are rational. Combining them with the retained strict interval $1.353893 < C_\partial < 1.354751$ proves AK18 by integer cross multiplication. The included script records the exact fractions. This numerical certificate does not strengthen the inherited analytic status of that C_∂ interval.

At the same concrete activation and with the original $r = k$ frame, AK11-12 therefore give the two evaluated complementary receivers

$$\begin{aligned} \mathcal{R}(\log \det H_K + \log \det H_{\text{rem},k}) &= \mathcal{R} \log \det G - [C_\partial - 2\log(4/\pi)]kq + o(kq), \\ \mathcal{R}(\Delta_K + \Delta_{\text{rem},k}) &= \mathcal{R}\Delta_E - 2\log(4/\pi)kq + o(kq). \end{aligned} \quad (\text{AK19})$$

All activated terms in AK19 are evaluated at $\alpha_k(\zeta_\diamond)$, and every Δ is relative to its own original unactivated metric. These formulas follow by substituting $C_{\text{inv}}(\zeta_\diamond) = C_\partial - 2\log(4/\pi)$ into the exact determinant factors. No extra term is adjoined to the arithmetic action.

7. A larger controlled quotient for the original kernel calculation

The estimate AK6 has $\min(m, r)$, not a mandatory factor r . Thus every admitted $r \log(q+2) = o(q)$ preserves the original kernel determinant through $o(kq)$, because $m = 8k - 16$ and $E_{r,N} = o(q)$. A specific growing choice supplied by this bound is

$$\begin{aligned} r_k &= \left\lfloor \frac{q}{\log^2(q+2)} \right\rfloor, \\ \sup_{0 \leq \alpha \leq 1} \left| \mathcal{R} \log \det H_{K,N}(\alpha) - \mathcal{R} \log \det \hat{H}_{K,r_k,N}(\alpha) \right| \\ &= O_{h,A} \left(k^2 \log^2(q+2) + \frac{Jkq}{\log(q+2)} \right) = o(kq). \end{aligned} \quad (\text{AK20})$$

Indeed $r_k \log(q+2) \leq q/\log(q+2) = o(q)$, so the original finite guards eventually hold, and substitution into the proved $E_{r,N}$ bound gives the displayed error. The exact shifted source in AK7-9 realizes this quotient for every activation. Its retained kernel dimension is still m ; all r_k inverse-power classes are removed by the actual quotient map. The four original cutoffs and source spaces are unchanged. The determinant of this remaining kernel is still the next value to evaluate; no numerical allocation follows merely from its unchanged dimension.

Proof sources and human sources

- IQR1-6 and IQR2's exact coefficient identification.
- IFH1-10: original observation, its finite inverse-power metric and exact determinant receiver.
- J3 and the complete joint-source minimum.
- TC1-29: the actual adjacent two-column source and complete normal-source covariance.
- Human analytic inputs carried by these original-source proofs: R. A. Askey and R. Roy, DLMF Chapter5, especially 5.8.E3; T. H. Koornwinder, R. Wong, R. Koekoek and R. F. Swarttouw, DLMF Chapter18, especially 18.22.E8 and 18.23.E7. Their original equation TeX is retained. The finite matrix identities AK4-17 are proved here; the human sources are credited for the Gamma and polynomial inputs rather than for these new receiving identities.

Part II

Full harmonic comparison and critical activation

Full-space harmonic control of the original critical activation

21 September 2026. Independent derivation of the received critical-activation equations (10)–(26), with the original physical coordinates, complete source covariance, and complete observed minimum retained. The new finite result is a full-space covariance comparison and its harmonic restriction to the original inverse-power frame. Its asymptotic high endpoint uses the already established IFH8-10 monic-profile and zero-law theorem at that theorem’s stated scope.

All constants in the received finite comparison are valid on the explicit IVO guards below. Two useful strengthenings are proved here: the lattice-product error is $O_{\delta,\gamma}(k)$, and the clipped critical law is uniform on the entire physical activation range, expressed as $\zeta \geq -2\log(q/k)$, not merely on fixed compact sets of ζ .

1. Source identity, inherited finite inputs, and exact guards

The complete received manuscript `SUPPLIED_CRITICAL_ACTIVATION.md` was read, SHA-256 5925D871EC57D58F3E993D97240F5061DE0D4726D3C4B3581481999929B6DA5C. The entire second attachment `SUPPLIED_CODE_AND_PROOF.md` was also read, SHA-256 63F17FA57E656542C108048942750D7C992ECAD55046FBDAAC99888E89C6AD3A. Its code and reported receipts are provenance, not an additional theorem or an execution performed here. In particular its preceding activation-source hash 6fc047f2513c6c61e0568e47bda23ccb5ccc4394e7704bb7f3f4e5aa9d871e29 agrees with the retained original activation proof previously read.

The web attachment’s manifest identifies its inverse-observation input as 28,714 bytes, SHA-256 bfdae1dd6fc2b99a045277141c3c0cdaf075cc3d7cc8408ed2eaf29f3e0f5e4b. It does not byte-match the retained `../observation_inverse_power_intake_20260921/INVERSE_POWER_OBSERVATION_SUPPLIED.md`, which is 29,503 bytes, SHA-256 78B06ADF8AA4065DACD76F901515D8FE88C736672107F4D6859892F3CE5FCD19; the retained supplied file in `publication_012` has this same latter size and hash. Consequently the manifest is not evidence of identical input versions. The verified IVO and IFH proofs listed below are the actual accepted providers used in this derivation.

The mathematical input files actually inspected are:

- `../publication_012/INDEPENDENT_FINITE_OBSERVATION.tex`, complete IVO1–26, SHA-256 D9889E4493C966DCD712E9A10840004794D4D41A79A91C8C9393CE15D7AC2E1E. Its pinned complete proof supplies IVO3, IVO9–11 and IVO24–26.
- `../INVERSE_FLAG_AND_RELATIVE_HEAT.tex`, IFH2–10, lines 41–190, SHA-256 468C92EE4224A3659BE44F122EFB9FBFC0C2A469624FF5DA6D16280AA069A6CA. The pinned IFH proof identifies the finite scalar sandwich and the inherited high-endpoint coefficient.
- `../verified_public_providers/NATIVE_GAUSSIAN_TRANSFER.tex`, NG1–3, lines 27–75, SHA-256 8A9FD3921F4A8667A88BBB5896AB7D5A19E5BE043FCB9B79BB883858888B72CA; NG2 in its pinned original source.

- The complete joint-source proof J3, lines 634–643 of `../../../../JOINT_MINIMUM_COMPLETE.tex`, previously read together with its physical section and full normal-source maps J5–J8. SHA-256 70DC576A470E1FDC15AFE7C7CA49EFCDD569BF1EA2A429094D0E19B97EBCD416. Its complete source is retained with the uniform-source-family edition.
- R. A. Askey and R. Roy’s DLMF Gamma product 5.8.3, original TeX at `../kernel_derivation/graph_primary_sources/DLMF_5_8_E3_Gamma_product_original.tex`, was read directly. SHA-256 83583AB37D82A6DE75A23F3183890BCE04A340AD4045A706993C2038EED1A34E. The Gamma bounds in Section 6 below are explicitly derived from it. IVO retains the original Meixner–Pollaczek conventions of T. H. Koornwinder, R. Wong, R. Koekoek and R. F. Swarttouw, with W. P. Reinhardt, DLMF 18.22.8 and 18.23.7; no replacement measure is used here.

Keep the original simple-quartet domain

$$\begin{aligned} 0 < \delta < \frac{1}{2}, \quad \gamma > 2, \quad k \equiv 1 \pmod{4}, \quad q = (k+1)^2, \\ S = k/2 + iy, \quad R_0 = \sqrt{\delta^2 + \gamma^2}, \quad R = kR_0, \end{aligned} \quad (\text{HAR1})$$

and the complete polynomial

$$Q(y) = \prod_{a,b=0}^k (y - \omega_{ab}), \quad \omega_{ab} = (2b - k)\gamma - i(2a - k)\delta, \quad E = \mathbb{C}[y]/(Q). \quad (\text{HAR2})$$

All roots are distinct and nonzero. The polynomial is real, even, and positive on the real axis. The native source remains $d\mu = w_h^{*k}(y) dy$, and the reference measure remains

$$d\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy, \quad M_\sigma = \sqrt{2\pi}. \quad (\text{HAR3})$$

Keep the full scalar and joint attained metrics G_N, G_N^J , and the same physical tensor section at all cutoffs. The source activation is exactly

$$C_N(\alpha) = G_N(\alpha)^{-1} = (1 - \alpha)G_N^{-1} + \alpha(G_N^J)^{-1}, \quad 0 \leq \alpha \leq 1. \quad (\text{HAR4})$$

The full-source comparison, in the same value coordinates, is

$$\mathbf{L}_k^{-1}T_k \preceq G_N^J \preceq \mathbf{B}_kT_k, \quad G_N^J \preceq G_N, \quad (\text{HAR5})$$

where T_k is the complete fixed tensor-value metric. In root-value coordinates its diagonal entries are the original ordered-tuple weights:

$$T_k = \text{diag}(W_\omega), \quad W_\omega \geq a_{\min}^k > 0, \quad \sum_{\omega} W_\omega = A_w^k. \quad (\text{HAR6})$$

The inequalities HAR5 follow from J3 before taking the minimum over the full source fibre on the lower side, and from the actual tensor section on the upper side. The already proved A6–7 bounds give individual logarithmic costs $O_h(k \log^2(q+2))$ for $\mathbf{L}_k, \mathbf{B}_k$; both use the same section and a degree allowance covering the four original cutoffs. A bound only on their product would not, by itself, bound either separate constant without further information. Here the retained source construction supplies that information.

For the onto original observation $\Lambda : E \rightarrow B$, define $Q_{B,N}(\alpha) = (\Lambda C_N(\alpha) \Lambda^*)^{-1}$. Use the literal inverse-power frame $U_r d = \sum_{j=1}^r d_j [y^{-j}]$, and its original source and observed Grams

$$H_{r,N}^E(\alpha) = U_r^* G_N(\alpha) U_r, \quad H_{r,N}^B(\alpha) = U_r^* \Lambda^* Q_{B,N}(\alpha) \Lambda U_r. \quad (\text{HAR7})$$

Set

$$n_0 = 2 \left\lfloor \frac{N+1-q}{2} \right\rfloor, \quad K_N = K_{Q,n_0}(0,0), \quad S_N = \frac{u_k}{|Q(0)|^2 K_N}. \quad (\text{HAR8})$$

The kernel is for the complete measure $Q^2 d\sigma$, and u_k is the unchanged NG2 constant. IVO/IFH gives the finite sandwich

$$a_{r,N} S_N I_r \preceq H_{r,N}^B(0) \preceq H_{r,N}^E(0) \preceq b_{r,N} S_N I_r, \quad (\text{HAR9})$$

with exactly

$$a_{r,N} = \frac{\ell_{k,N}}{u_k} \left(\frac{\mathcal{L}_* \mathcal{R}_*}{2\mathcal{F}_N \mathcal{B}^{Jr}} \right)^2, \quad b_{r,N} = \mathcal{U}_{r,N}^2. \quad (\text{HAR10})$$

Here each conductor constant is the complete displayed IVO constant, for an actual nonzero selected conductor coefficient. The finite domain retained is IVO24:

$$q_- = (k-7)^2 \geq 16, \quad R \leq 2^{-14} q_-, \quad n_- + 2 \leq 2q_-, \quad \mathcal{L}_* \mathcal{R}_* T_N^{\text{lb}} \geq 2\mathcal{A}, \quad n_- = N - v + Jr - q_-. \quad (\text{HAR11})$$

Also retain $1 \leq r \leq q$, the actual conductor identity through the full observation, and its proved five-orbit period domain. Under the eventual regime $r \log(q+2) = o(q)$, every guard in HAR11 holds for the same fixed original data, as proved in IVO24-26. In particular

$$\log^+(a_{r,N}^{-1}) + \log^+ b_{r,N} = O_{h,A}((k+Jr) \log(q+Jr+2)). \quad (\text{HAR12})$$

No part of the argument below asserts these eventual guards at the small lattice sizes used only for exact algebraic checks.

2. The exact lattice cardinal bound

Write $D_{ab} = |Q'(\omega_{ab})|$. For a fixed row $m \neq a$, the factors in $D_{a,b+1}^2 / D_{ab}^2$ cancel successively along that row; only its two end factors remain. In row $m = a$, the omitted self-factor gives the horizontal product $(2\gamma)^{2k} (b!)^2 ((k-b)!)^2$, whose adjacent ratio is $(b+1)^2 / (k-b)^2$. Combining all rows proves exactly

$$\frac{D_{a,b+1}^2}{D_{ab}^2} = \prod_{m=0}^k \frac{\delta^2(a-m)^2 + \gamma^2(b+1)^2}{\delta^2(a-m)^2 + \gamma^2(k-b)^2}. \quad (\text{HAR13})$$

All factors are less than one if $b < (k-1)/2$, equal to one if $b = (k-1)/2$, and greater than one afterwards. Interchanging the two coordinate directions proves the same statement in a . Hence precisely the four central roots minimize D_{ab} , with equal minima. Each has modulus R_0 .

Choose a central root ω_c . A shell of index displacement $j = \max(|a - a_c|, |b - b_c|)$ contains at most $8j$ roots, and every distance in that shell is at least $2\delta j$. The largest possible j is $(k+1)/2$, in particular at most k . Since $|\omega| \leq |\omega - \omega_c| + R_0$, the inequality $\log(1+x) \leq x$ gives

$$\begin{aligned} \log \frac{|Q(0)|}{R_0|Q'(\omega_c)|} &= \sum_{\omega \neq \omega_c} \log \frac{|\omega|}{|\omega - \omega_c|} \\ &\leq R_0 \sum_{\omega \neq \omega_c} \frac{1}{|\omega - \omega_c|} \leq \frac{2(k+1)R_0}{\delta} \leq \frac{4kR_0}{\delta}. \end{aligned} \quad (\text{HAR14})$$

The exact minimum property of the central derivative now proves the supplied constant

$$\boxed{\frac{|Q(0)|}{|Q'(\omega)|} \leq R_0 e^{4kR_0/\delta} \quad \text{for every original root.}} \quad (\text{HAR15})$$

The preceding line also supplies the smaller exponent $2(k+1)R_0/\delta$, if desired. HAR15 retains the received constant exactly.

3. Cardinal lifts and the complete trace comparison

The IVO hard gap has $\beta_q = 2^{-27}q^2$. HAR11 implies $R^2 \leq 2^{-28}q^2 \leq \beta_q/2$. Also $q \geq 16$ and $n_0 \leq q \leq 2q - 2$ at every original cutoff. Thus IVO11 applies to

$$H(y) = \frac{K_{Q,n_0}(y, 0)}{K_N}, \quad H(0) = 1, \quad \|QH\|_\sigma^2 = K_N^{-1}, \quad |H(z)| \geq e^{-n_0 R^2/\beta_q} \quad (|z| \leq R). \quad (\text{HAR16})$$

The last inequality follows from the paired zeros of the odd polynomial $yH(y)$: their nonzero squares are at least β_q , and $\log(1-u) \geq -2u$ for $0 \leq u \leq 1/2$. This is exactly the IVO11 constant, now with its required radius and degree guards checked.

For each original root use the actual polynomial

$$P_\omega(y) = \frac{Q(y)H(y)}{(y-\omega)Q'(\omega)H(\omega)}. \quad (\text{HAR17})$$

Its degree is at most $q+n_0-1 \leq N$, and its complete root values are $P_\omega(\omega') = \mathbf{1}_{\omega=\omega'}$. Thus it lifts the original cardinal class e_ω , without replacing its value coordinates. For real y , oddness of k gives $|\text{Im } \omega| \geq \delta$, so $|y - \omega| \geq \delta$. NG2, applied to this full degree- N polynomial, yields

$$\|e_\omega\|_{G_N}^2 \leq \frac{u_k e^{2n_0 R^2/\beta_q}}{\delta^2 |Q'(\omega)|^2 K_N}. \quad (\text{HAR18})$$

In the original root-value frame, e_ω is the corresponding coordinate vector and HAR6 is diagonal. Consequently

$$\text{Tr}(T_k^{-1}G_N) = \sum_{\omega} \frac{\|e_\omega\|_{G_N}^2}{W_\omega} \leq \Gamma_N S_N, \quad (\text{HAR19})$$

where the received explicit constant is

$$\boxed{\Gamma_N = \max \left(1, \frac{qR_0^2}{\delta^2 a_{\min}^k} \exp \left[\frac{8kR_0}{\delta} + \frac{2n_0 R^2}{\beta_q} \right] \right)}. \quad (\text{HAR20})$$

Changing back to the fixed polynomial frame changes both T_k and G_N by the same congruence, so the trace and the inequality are unchanged. The term $2n_0 R^2/\beta_q$ is bounded at fixed δ, γ , since $n_0 \leq q$ and $k^2/q \leq 1$. Therefore $\log \Gamma_N = O_h(k + \log q)$. This controls the full metric, including all cross terms; no diagonal replacement of G_N was made.

4. The full-space harmonic comparison and the original observation

For any positive T, G , the positive matrix $G^{1/2}T^{-1}G^{1/2}$ is at most its trace times the identity. Congruence by $G^{-1/2}$ proves

$$T^{-1} \preceq \text{Tr}(T^{-1}G)G^{-1}. \quad (\text{HAR21})$$

Invert the lower half of HAR5 and apply HAR19–21 to obtain

$$(G_N^J)^{-1} \preceq \mathbf{L}_k T_k^{-1} \preceq \mathbf{L}_k \Gamma_N \mathbf{S}_N G_N^{-1}.$$

Substitution in the complete covariance HAR4 and inversion give the stronger full-space result

$$G_N(\alpha) \succeq \frac{G_N}{1 - \alpha + \alpha \mathbf{L}_k \Gamma_N \mathbf{S}_N}. \quad (\text{HAR22})$$

Applying the unchanged Λ, Λ^* to the covariance order before inversion gives, on the whole observation space,

$$Q_{B,N}(\alpha) \succeq \frac{Q_{B,N}(0)}{1 - \alpha + \alpha \mathbf{L}_k \Gamma_N \mathbf{S}_N}. \quad (\text{HAR23})$$

Restriction to $F = \Lambda U_r$ is exactly the received observed lower bound. This derivation never commutes a restriction with inversion.

The left inverse used in received (14) is also exact. With $H_0 = F^* Q_{B,N}(0) F > 0$, put $R_N = H_0^{-1} F^* Q_{B,N}(0)$. Multiplication gives $R_N F = I_r$ and $R_N Q_{B,N}(0)^{-1} R_N^* = H_0^{-1}$. For any positive Q_B , minimizing $x^* Q_B x$ under $R_N x = d$ gives the attained metric $(R_N Q_B^{-1} R_N^*)^{-1}$. The vector Fd is one admissible lift, so $F^* Q_B F \succeq (R_N Q_B^{-1} R_N^*)^{-1}$. This verifies that route as well, with the full target minimum retained.

The two upper covariance comparisons yield

$$H_{r,N}^E(\alpha) \preceq \frac{b_{r,N} \mathbf{S}_N}{1 - \alpha} I_r, \quad H_{r,N}^E(\alpha) \preceq \frac{\mathbf{B}_k}{\alpha} U_r^* T_k U_r, \quad (\text{HAR24})$$

where the corresponding inequality is omitted if its denominator is zero. For every coefficient vector d , the original root-value norm satisfies

$$\begin{aligned} d^* U_r^* T_k U_r d &= \sum_{\omega} W_{\omega} \left| \sum_{j=1}^r d_j \omega^{-j} \right|^2 \\ &\leq \|d\|^2 \sum_{\omega} W_{\omega} \sum_{j=1}^r |\omega|^{-2j} \leq \frac{A_w^k}{R_0^2 - 1} \|d\|^2. \end{aligned} \quad (\text{HAR25})$$

Here every root satisfies $|\omega| \geq R_0 > 2$, and the two displayed coefficient sums include all inverse powers and all original root weights.

Define, with the received notation,

$$h_N(\alpha) = \frac{\mathbf{S}_N}{1 - \alpha + \alpha \mathbf{S}_N}, \quad L_{r,N} = \frac{a_{r,N}}{\max(1, \mathbf{L}_k \Gamma_N)}, \quad U_{r,N} = b_{r,N} + \frac{\mathbf{B}_k A_w^k}{R_0^2 - 1}. \quad (\text{HAR26})$$

For positive b, d, s and $0 < \alpha < 1$,

$$\min \left(\frac{bs}{1 - \alpha}, \frac{d}{\alpha} \right) \leq (b + d) \frac{s}{1 - \alpha + \alpha s}. \quad (\text{HAR27})$$

To verify it, if the first fraction is smaller, its ratio to the rightmost $s/(1 - \alpha + \alpha s)$ is $b + b\alpha s/(1 - \alpha) \leq b + d$. If the second is smaller the same calculation gives $d + d(1 - \alpha)/(\alpha s) \leq d + b$. At the endpoints retain the finite fraction. Also $1 - \alpha + \alpha cs \leq \max(1, c)(1 - \alpha + \alpha s)$. Thus HAR9, HAR23-27 give

$$\boxed{L_{r,N}h_N(\alpha)I_r \preceq H_{r,N}^B(\alpha) \preceq H_{r,N}^E(\alpha) \preceq U_{r,N}h_N(\alpha)I_r \quad (0 \leq \alpha \leq 1).} \quad (\text{HAR28})$$

The middle inequality is the original observed minimum over its full fibre, which includes the original vector as one lift.

Set $E_{r,N} = \max(0, -\log L_{r,N}, \log U_{r,N})$. The individual source constants, HAR12, HAR20, and the fixed masses in HAR6 show

$$E_{r,N} = O_{h,A} \left(k \log^2(q+2) + (k+Jr) \log(q+Jr+2) \right) = o(q) \quad (\text{HAR29})$$

under $r \log(q+2) = o(q)$. Indeed J is fixed, $r = o(q/\log q)$, and $k \log^2 q = o(q)$. No assertion about a smaller source rank is used. Every original Gram eigenvalue therefore obeys

$$|\log \lambda_j(H_{r,N}^X(\alpha)) - \log h_N(\alpha)| \leq E_{r,N}, \quad X = E, B, \quad (\text{HAR30})$$

and $|\log \det H_{r,N}^X(\alpha) - r \log h_N(\alpha)| \leq r E_{r,N}$. The full relative observation bound is $H^B \succeq e^{-2E_{r,N}} H^E$, uniformly in α . Its kernel consequences are recorded by the receiving kernel derivation; no initial angle has been discarded here.

At the four original cutoffs, $n_0 = 0, 0, q, q$, since q is even. Hence there are exactly two values S_L, S_H . With the unchanged four signs,

$$\boxed{\left| \mathcal{R} \log \det H_{r,N}^X(\alpha) - 2r \log \frac{S_L(1 - \alpha + \alpha S_H)}{S_H(1 - \alpha + \alpha S_L)} \right| \leq r \sum_N E_{r,N}.} \quad (\text{HAR31})$$

This is a finite all-activation enclosure using the two original scalar relation kernels.

5. Exact rectangular integral and a controlled lattice product

Write

$$I_{\square} = \frac{1}{4\delta\gamma} \int_{-\delta}^{\delta} \int_{-\gamma}^{\gamma} \log(x^2 + y^2) dy dx. \quad (\text{HAR32})$$

The logarithmic singularity is integrable. On the positive rectangle, apply $\text{div}((x, y) \log(x^2 + y^2)) = 2 \log(x^2 + y^2) + 2$ after removing a small quarter-disc; its circular boundary term tends to zero. The two outer boundary integrals use $\int_0^b \log(a^2 + t^2) dt = b \log(a^2 + b^2) - 2b + 2a \arctan(b/a)$. Dividing by the rectangle area gives

$$I_{\square} = \log(\delta^2 + \gamma^2) - 3 + \frac{\delta}{\gamma} \arctan \frac{\gamma}{\delta} + \frac{\gamma}{\delta} \arctan \frac{\delta}{\gamma}. \quad (\text{HAR33})$$

Here is a finite error bound for the exact lattice rather than an informal Riemann-sum argument. Put $n = k + 1$, and let $x_a = (2a - k)/n$, $0 \leq a \leq k$, be the midpoint grid on $[-1, 1]$. For $f(x, y) = \log(\delta^2 x^2 + \gamma^2 y^2)$, the original product is exactly

$$\log |Q(0)|^2 = 2q \log n + \sum_{a,b=0}^k f(x_a, x_b). \quad (\text{HAR34})$$

The four central cells have centres $(\pm 1/n, \pm 1/n)$. Their centre value is $-2 \log n + \log R_0^2$, while their cell average is $-2 \log n + \log 4 + I_\square$, by the exact scaling of HAR32. Their total error is at most $4 |\log R_0^2 - \log 4 - I_\square|$.

The shell with maximum centre-coordinate magnitude $(2j+1)/n$, $j \geq 1$, has $8j+4 \leq 12j$ cells. Every point in each such cell has Euclidean distance at least $2j/n$ from zero. Since $\|\nabla f\| \leq 2\gamma^2/(\delta^2 \sqrt{x^2 + y^2})$, and the distance from the centre to a point of its cell is at most $\sqrt{2}/n$, each centre-versus-cell-average error is at most $\sqrt{2}\gamma^2/(\delta^2 j)$. Summing at most $n/2$ shells proves

$$|\log |Q(0)|^2 - 2q \log k - qI_\square| \leq 2q \log \frac{k+1}{k} + \frac{6\sqrt{2}\gamma^2}{\delta^2} (k+1) + 4 |\log R_0^2 - \log 4 - I_\square|. \quad (\text{HAR35})$$

The right side is $O_{\delta,\gamma}(k)$, which also proves the received $O_h(k \log(k+2))$ bound. The four central cells and every original root are retained.

6. The low Gamma monic norm and the unchanged source mass

Let $\sigma_0 = \Gamma(1/4)^2/(2\pi)$, $a = 1/4$, and $u = |y|/2$. The original-author Gamma product gives

$$\frac{\sigma(y)}{\sigma_0} = \prod_{j=0}^{\infty} \left(1 + \frac{u^2}{(j+a)^2} \right)^{-1}.$$

The positive summand $g(x) = \log(1+u^2/(x+a)^2)$ decreases, so its sum lies between $\int_0^\infty g$ and $g(0) + \int_0^\infty g$. Direct integration gives $\int_0^\infty g = 2u \arctan(u/a) - a \log(1+u^2/a^2)$. Substitution and $0 \leq |y| \arctan(1/(2|y|)) \leq 1/2$ prove

$$\frac{\sigma_0 e^{-\pi|y|/2}}{1+4y^2} \leq \sigma(y) \leq \sigma_0 e^{1/2} (1+4y^2)^{1/4} e^{-\pi|y|/2}. \quad (\text{HAR36})$$

For the lower side the integral comparison first gives the stronger denominator $(1+4y^2)^{3/4}$, which implies the displayed bound. At $y = 0$ the formulas follow by continuity. Thus both received Gamma envelope constants are verified.

Let $\nu_0^\Gamma = \int |Q(y)|^2 d\sigma(y)$, retain $R = kR_0$, and put $c = \pi/2$, $x_0 = 2q/c = 4q/\pi$. The full root product gives $|Q(y)| \leq (|y| + R)^q$ everywhere, and $|Q(y)| \geq (|y| - R)^q$ when $|y| \geq R + 1$. HAR36 and $(1+4y^2)^{1/4} \leq 1+2|y|$ give the explicit upper bound

$$\nu_0^\Gamma \leq 2\sigma_0 e^{1/2+cR} \left[\frac{\Gamma(2q+1)}{c^{2q+1}} + \frac{2\Gamma(2q+2)}{c^{2q+2}} \right]. \quad (\text{HAR37})$$

For a lower bound integrate only $y \in [R+x_0, R+x_0+1]$, where $x_0 > 1$. No root is removed from Q ; the discarded integration region is used only for a lower bound. This gives

$$\nu_0^\Gamma \geq \frac{\sigma_0 x_0^{2q} e^{-c(R+x_0+1)}}{1+4(R+x_0+1)^2}. \quad (\text{HAR38})$$

The factorial estimate $\log(n!) = n \log n - n + O(\log(n+2))$, obtainable by bounding the sum of $\log j$ between its two adjacent integrals, now yields

$$\log \nu_0^\Gamma = 2q \log q + (2 \log(4/\pi) - 2)q + O_h(k + \log q). \quad (\text{HAR39})$$

It remains necessary to bound the actual u_k , rather than only its ratio with the lower source constant. Put $m_h = \int_{\mathbb{R}} w_h(y) dy > 0$. The constant polynomial in NG2 gives $\ell_{k,N} \leq m_h^k / M_\sigma \leq u_k$. The exact retained NG2 constants have $u_k = A_k$, $\ell_{k,N} =$

$a_k/[1 + 4(N + M)^2]^M$, $M = 25$ on this simple-quartet domain, and $\log(u_k/\ell_{k,N}) = O_h(k + \log(N + 1))$. Therefore

$$\log \frac{m_h^k}{M_\sigma} \leq \log u_k \leq \log \frac{m_h^k}{M_\sigma} + \log \frac{u_k}{\ell_{k,N}}, \quad |\log u_k| = O_h(k + \log q). \quad (\text{HAR40})$$

The second inequality uses both the constant-polynomial bounds and the comparison ratio. The constant-polynomial test alone would be insufficient. No source mass has been divided out or changed.

Since $K_{Q,0}(0,0) = 1/\nu_0^\Gamma$, equations HAR8, HAR35, HAR39–40 prove

$$\log S_L = 2q\ell_k + \beta_L q + O_h(k + \log q), \quad \ell_k = \log(q/k), \quad a_0 = \log(4/\pi), \quad \beta_L = 2a_0 - 2 - I_\square. \quad (\text{HAR41})$$

This lower endpoint has been derived directly from the full original lattice and Gamma product.

7. The high endpoint and the clipped coefficient

At the high endpoint, the exact IFH8 identity retains the monic polynomials V_j and squared norms ν_j^Γ of $Q^2 d\sigma$:

$$\log \frac{K_{Q,q}(0,0)}{K_{Q,0}(0,0)} = 2 \log |V_q(0)| + \log \nu_0^\Gamma - \log \nu_q^\Gamma + \log \Xi_q. \quad (\text{HAR42})$$

The original parity, Christoffel-Darboux formula, and finite hard gap give $1 \leq \Xi_q \leq 4(2q + 2)^2/\beta_q$, so $\log \Xi_q = O(1)$. IFH8–10 applies the already retained full monic profile and zero law to the other terms, in their original source scale, and proves

$$\log \frac{K_{Q,q}(0,0)}{K_{Q,0}(0,0)} = \frac{\mathfrak{C}}{2}q + o_h(q), \quad \mathfrak{C} = 4[\psi(0) - \psi(1) - J(1) - 1]. \quad (\text{HAR43})$$

This is the precise analytic source theorem imported for the high endpoint. Its original finite-zero gap permits the logarithmic test function in the zero law. The inherited positive interval $1.353893 < \mathfrak{C} < 1.354751$ is not a new numerical computation in this note. Therefore

$$\log S_H = 2q\ell_k + \beta_H q + o_h(q), \quad \beta_H = \beta_L - \mathfrak{C}/2 < \beta_L. \quad (\text{HAR44})$$

In particular both S_L, S_H exceed 1 eventually.

For any $s \geq 1$, $0 \leq \alpha \leq 1$, the exact denominator obeys $\max(1, \alpha s) \leq 1 - \alpha + \alpha s \leq 2 \max(1, \alpha s)$. Taking logarithms proves

$$\min(\log s, -\log \alpha) - \log 2 \leq \log \frac{s}{1 - \alpha + \alpha s} \leq \min(\log s, -\log \alpha), \quad (\text{HAR45})$$

with $-\log 0 = +\infty$. This proves the constant $\log 2$ uniformly, including the activation endpoints.

Let $\alpha_k(\zeta) = \exp[-2q\ell_k - \zeta q]$. Its exact physical domain is $\zeta \geq -2\ell_k$; $\alpha = 0$ is its endpoint $\zeta = +\infty$. Define the endpoint error

$$\epsilon_k^{\text{end}} = \max_{T=L,H} |\log S_T - 2q\ell_k - \beta_T q| = o_h(q). \quad (\text{HAR46})$$

Since $x \mapsto \min(x, c)$ is 1-Lipschitz, HAR30 and HAR45 imply the finite uniform estimate

$$\boxed{|\log \lambda_j(H_{r,N}^X(\alpha_k(\zeta))) - 2q\ell_k - q \min(\beta_{L/H}, \zeta)| \leq E_{r,N} + \epsilon_k^{\text{end}} + \log 2.} \quad (\text{HAR47})$$

It holds for every eigenvalue, $X = E, B$, and every physical $\zeta \geq -2\ell_k$, once $S_L, S_H \geq 1$. Thus the limiting error is uniform on that entire moving domain, not only on its fixed compact subsets. It applies to every sequence of physical activations through this exact finite inequality.

Summing the r eigenvalue logarithms with the unchanged four cutoff signs yields

$$\boxed{\mathcal{R} \log \det H_{r,N}^X(\alpha_k(\zeta)) = rq\mathfrak{C}_{\text{inv}}(\zeta) + o_{h,A}(rq), \quad \mathfrak{C}_{\text{inv}}(\zeta) = 2[\min(\beta_L, \zeta) - \min(\beta_H, \zeta)].} \quad (\text{HAR48})$$

An explicit error is $r \sum_N E_{r,N} + 4r\epsilon_k^{\text{end}} + 4r \log 2$, independent of ζ . Because $\mathfrak{C} > 0$, this coefficient equals 0 on $\zeta \leq \beta_H$, equals $2(\zeta - \beta_H)$ between the two thresholds, and equals $\mathfrak{C}$ on $\zeta \geq \beta_L$. The theorem covers $r = k$ and every admitted $r \log(q+2) = o(q)$.

Finally $\log(x^2 + y^2) > \log(y^2)$ almost everywhere in the positive rectangle, so $I_{\square} > \gamma^{-1} \int_0^\gamma \log(y^2) dy = 2 \log \gamma - 2$. Hence

$$\beta_L < 2 \log \frac{4}{\pi\gamma} < 0. \quad (\text{HAR49})$$

For the coarser source activation $e^{-2bq\ell_k}$, $b \geq 0$, the exact substitution is $\zeta_k = 2(b-1)\ell_k$. The uniform result HAR48 therefore gives zero return coefficient for fixed $0 \leq b < 1$, and $\mathfrak{C}$ for fixed $b > 1$. At $b = 1$, $\zeta = 0 > \beta_L$ by HAR49, so the latter value includes that endpoint. This is an application of a proved uniform finite estimate, not a moving-parameter substitution into a merely pointwise theorem.

8. Exact finite checks and their limits

`check_harmonic.py` was written independently in this intake and executed with Python and SymPy. Its `HARMONIC_EXACT_CHECKS.json` records **358 passing exact checks**. They include both coordinate-direction derivative-product identities on the actual rectangular coordinates $\delta = 1/4, \gamma = 3$, at $k = 1, 5, 9$; the exact four central derivative minima; every shell-count and shell-distance inequality used above; and a noncommuting complex-rational source fixture with all source and observation cross terms present.

For that fixture every principal minor is checked exactly for the full-space trace order, the joint metric enclosure, the canonical observed left inverse, and the harmonic source and observed inequalities at activations $0, 1/13, 1/2, 1$. Its coefficient frame consists of the literal inverse powers of the four roots $\pm 3 \pm i/4$. The scalar clipping check uses its equivalent rational denominator bounds, including both activation endpoints.

These checks establish exact finite algebra for their declared fixtures. They do not evaluate native arithmetic moments or an actual period, and the small lattice tests do not claim the eventual hard-gap guards. The asymptotic high coefficient retains the exact IFH/H-profile source status identified in HAR42–44. The complete received arithmetic-action and heat statements outside equations (10)–(26) are not claimed to have been independently proved by this harmonic derivation.

Part III

Original arithmetic action and all singular directions

The original inverse and observed arithmetic action at critical activation

Independent mathematical derivation, 21 September 2026. This file proves the action conclusions in supplied equations (27)–(48). It supplies the exact covariance coisometry and the missing quantitative shear calculation. The original metric, polynomial minimum section, physical coordinate, observation, conductor coefficients, and period stratum are retained.

1. Sources and precise shared input

The supplied critical-activation text is retained as **SUPPLIED_CRITICAL_ACTIVATION.md** in this directory. Its equation (17), together with the logarithmic costs in (18), is proved independently as **HAR28-31** in the companion **HARMONIC_PROOFS.md**. That proof's **HAR41-48** evaluates the scalar endpoints and their clipped law. These precise provider sections were read for the present derivation. The present proof uses the harmonic enclosure for the inverse norm, observation fraction, action-source injectivity, common-subspace determinant cancellation, and action shear estimate. No action or singular-value conclusion is included in this input.

The finite inequalities retain every activation $0 \leq \alpha \leq 1$. The critical limits in Sections 4 and 7–9 take ζ, b fixed, or uniformly in bounded parameter ranges; the heat conclusions additionally require the displayed strict gaps to be bounded away from zero. They do not claim these critical limits uniformly over arbitrary moving physical activation parameters. For example $\zeta_k = -2\log(q/k)$ gives $\alpha_k = 1$ and $h_N(1) = 1$; neither a superexponentially small singular value nor a vanishing shear follows there from the critical argument. The finite estimates still apply there.

Earlier source files inspected at the following exact proof locators:

- The original coordinate conductor, full-fibre minimum, source comparison and anchored division, **IVO1-6**, in the pinned independent finite observation proof. Its finite guards and inverse-power comparison are **IVO24-26**, in the same verified file. Local TeX SHA-256: d9889e4493c966dcd712e9a10840004794d4d41a79a91c8c9393ce15d7ac2e1e, equal to the retained full-byte publication receipt.
- The original inverse decomposition and Gamma evaluation/anchored-division estimates, **S11-16**, in the pinned smallest-singular proof. Local TeX SHA-256: c7dd360325dc475b82cabbc65f2dc01d5c67711b85951a1837cd3e7f52415bbb, equal to that receipt.
- The original observed isometry **IFH1**, full flag minima **IFH2-7**, and relative measured-heat argument **IFH11-13**, in the pinned original inverse-flag proof. Local TeX SHA-256: 468c92ee4224a3659be44f122efb9fbfc0c2a469624ff5da6d16280aa069a6ca.
- The retained activation source **A23-25** contains the earlier covariance coisometry for the outgoing action; **A38** retains the same full spectral-polynomial re-

ceiver. Those sections were read, and the construction is proved again for the inverse below.

- The received dual spectral-pencil proof **D1-2** contains the determinant formula. Its whole determinant calculation is reproduced in Section 8 below.

The source-reading ledger was used before these files were opened. The continuation's claimed inverse-observation byte identity differs from our retained accepted supplied proof: the latter has 29,503 bytes and SHA-256 78b06adf8aa4065dacd76f901515d8fe88c736672107f4d6859892f3ce5fcd19. We use the actual accepted IFH/IVO providers, not an unseen claimed-identical file. No PDF was read.

The human sources retained by IVO include DLMF Chapter 18 by T. H. Koornwinder, R. Wong, R. Koekoek, R. F. Swarttouw, with W. P. Reinhardt, especially Meixner-Pollaczek generating function 18.23.E7 and recurrence 18.22.E8. The finite consequences needed here are proved in IVO5-6. The original Gamma source mass remains $\sqrt{2\pi}$.

Keep

$$q = (k+1)^2, \quad k \equiv 1 \pmod{4}, \quad 0 < \delta < 1/2, \quad \gamma > 2, \quad R_0 = (\delta^2 + \gamma^2)^{1/2},$$

$$Q_k(y) = \prod_{a,b=0}^k [y - (2b-k)\gamma + i(2a-k)\delta], \quad E_k = \mathbb{C}[y]/(Q_k), \quad M[p] = [yp].$$

The original coordinate is $S = k/2 + iy$. Since $Q_k(0) \neq 0$, M is invertible; every root has modulus at least R_0 .

Fix a cutoff N . Put $G_0 = G_N$, $G_1 = G_N^J$, and

$$G_\alpha^{-1} = (1 - \alpha)G_0^{-1} + \alpha G_1^{-1}, \quad 0 \leq \alpha \leq 1. \quad (\text{ACT1})$$

The joint source contains the canonical source, so $G_1 \preceq G_0$. Consequently $G_1 \preceq G_\alpha \preceq G_0$; for $\alpha > 0$, $G_\alpha \preceq \alpha^{-1}G_1$.

Retain

$$Q_\alpha = (\Lambda G_\alpha^{-1} \Lambda^*)^{-1}, \quad L_\alpha = G_\alpha^{-1} \Lambda^* Q_\alpha, \quad P_{B,\alpha} = L_\alpha \Lambda.$$

Direct multiplication gives $\Lambda L_\alpha = I$, $L_\alpha^* G_\alpha L_\alpha = Q_\alpha$, and $\Lambda = L_\alpha^\dagger$; $P_{B,\alpha}$ is the original orthogonal observed projection.

For $U_d z = \sum_{j=1}^d z_j [y^{-j}]$, write $H_{d,\alpha}^E = U_d^* G_\alpha U_d$ and $H_{d,\alpha}^B = U_d^* \Lambda^* Q_\alpha \Lambda U_d$. The shared harmonic lemma states

$$e^{-E_{d,N}} h_N(\alpha) I_d \preceq H_{d,\alpha}^B \preceq H_{d,\alpha}^E \preceq e^{E_{d,N}} h_N(\alpha) I_d, \quad h_N(\alpha) = \frac{S_N}{1 - \alpha + \alpha S_N}, \quad (\text{ACT2})$$

where $E_{d,N} = o(q)$ for $d \log(q+2) = o(q)$. It follows from the explicit $L_{d,N}, U_{d,N}$ in supplied (17); no metric is rescaled.

Every use retains the IVO guards at the enclosing rank d : $d \leq q$, $q_- \geq 16$, $R_k \leq 2^{-14} q_-$, $n_- + 2 \leq 2q_-$, and the complete absorption guard IVO24 with $n_- = N - v + Jd - q_-$, together with the shared harmonic lemma's guards. Sections 5-7 also retain $J(r-1) + 1 \leq q_-$, $r + v \leq q$, and $v \leq N$. These are actual finite domain restrictions and hold eventually in the stated growth regime.

2. Original polynomial inverse and covariance transport

Let $\hat{R}_N : E_k \rightarrow \mathcal{P}_N$ be the actual canonical scalar minimum section and $J_N : \mathcal{P}_N \rightarrow E_k$ its remainder map. The hat distinguishes this section from the coefficient left inverse also called R_N in supplied (14). Define

$$\mathcal{D}p = (p - p(0))/y, \quad B_N = J_N \mathcal{D} \hat{R}_N, \quad \ell_N x = (\hat{R}_N x)(0), \quad u = [y^{-1}].$$

The functional evaluates the specified polynomial representative, not the quotient algebra at zero. Multiplication gives $MB_Nx = x - \ell_N(x)[1]$ and $Mu = [1]$, hence

$$\boxed{T := M^{-1} = B_N + u\ell_N.} \quad (\text{ACT3})$$

The S11 vector $[(Q_k - Q_k(0))/y]$ equals $-Q_k(0)u$, so the original factor and matched row agree exactly.

IVO3 and IVO6 give

$$\|B_N\|_{G_0} \leq D_N := \mathfrak{d}_N \sqrt{u_k/\ell_{k,N}}, \quad \mathfrak{d}_N = 2\sqrt{2N+1} \left(\sum_{j=1}^N \frac{1}{j} \right) (1 + \sqrt{N+1}). \quad (\text{ACT4})$$

Indeed J_N is a contraction to the attained metric and $\hat{R}_N$ is an isometry; the two source comparisons apply to the actual polynomials before and after division. S14/IVO6 also gives

$$\|\ell_N\|_{G_0^{-1}} \leq F_N := \sqrt{\frac{N+1}{\ell_{k,N}\sqrt{2\pi}}}. \quad (\text{ACT5})$$

Here $K_{\sigma,N}(0,0) \leq (N+1)/\sqrt{2\pi}$, with original mass retained.

In $T_k = \text{diag}(W_\omega)$, T is diagonal with entries $1/\omega$; $\|T\|_{T_k} = R_0^{-1}$. Therefore

$$\|T\|_{G_1} \leq J_N^{\text{inv}} := \frac{\sqrt{\mathbf{L}_k \mathbf{B}_k}}{R_0}. \quad (\text{ACT6})$$

The superscript distinguishes this scalar bound from the remainder map.

Define

$$\mathcal{J}_\alpha : E_{G_0} \oplus E_{G_1} \rightarrow E_{G_\alpha}, \quad \mathcal{J}_\alpha(x_0, x_1) = \sqrt{1-\alpha}x_0 + \sqrt{\alpha}x_1.$$

Its actual adjoint is

$$\mathcal{J}_\alpha^\dagger z = (\sqrt{1-\alpha}G_0^{-1}G_\alpha z, \sqrt{\alpha}G_1^{-1}G_\alpha z).$$

Equation (ACT1) proves $\mathcal{J}_\alpha \mathcal{J}_\alpha^\dagger = I$. Thus it is a coisometry and its adjoint is an isometry. For every fixed coefficient operator A ,

$$A = \mathcal{J}_\alpha \text{diag}(A, A) \mathcal{J}_\alpha^\dagger, \quad \|A\|_{G_\alpha} \leq \max(\|A\|_{G_0}, \|A\|_{G_1}). \quad (\text{ACT7})$$

The equality retains factor order $A[(1-\alpha)G_0^{-1} + \alpha G_1^{-1}]G_\alpha = A$.

Apply the same map to $\text{diag}(B_N, T)$ and retain the canonical rank-one term separately:

$$\boxed{T = B_{\alpha,N} + (1-\alpha)u\ell_N G_0^{-1}G_\alpha,} \quad (\text{ACT8})$$

$$B_{\alpha,N} = (1-\alpha)B_N G_0^{-1}G_\alpha + \alpha T G_1^{-1}G_\alpha, \quad \|B_{\alpha,N}\|_{G_\alpha} \leq \overline{D}_N := \max(D_N, J_N^{\text{inv}}). \quad (\text{ACT9})$$

The moving dual row is exact. Neither selfadjointness nor nilpotence of these terms is asserted. The inherited finite source costs imply $\log^+ \overline{D}_N = O_h(k \log^2(q+2)) = o(q)$.

3. Inverse norm and every remaining singular direction

Here are explicit constants for supplied (28). Put

$$s = S_N, \quad b = b_{1,N}, \quad m_k = M_h(0)^k, \quad c_0 = (D_N/\sqrt{s} + \sqrt{b}F_N)^2, \quad C_0 = \max(c_0, (J_N^{\text{inv}})^2/s).$$

The canonical $r = 1$ bound gives $\|u\|_{G_0}^2 \leq bs$, so (ACT3) implies $\|T\|_{G_0}^2 \leq c_0 s$, and (ACT7) gives $\|T\|_{G_\alpha}^2 \leq C_0 s$. For $\alpha > 0$,

$$\|Tx\|_{G_\alpha}^2 \leq \alpha^{-1} \|Tx\|_{G_1}^2 \leq \alpha^{-1} (J_N^{\text{inv}})^2 \|x\|_{G_1}^2 \leq \alpha^{-1} (J_N^{\text{inv}})^2 \|x\|_{G_\alpha}^2.$$

Thus

$$\|T\|_{G_\alpha}^2 \leq \min(C_0 s, (J_N^{\text{inv}})^2/\alpha) \leq [C_0 + (J_N^{\text{inv}})^2] h_N(\alpha). \quad (\text{ACT10})$$

For the last step, the minimum z satisfies $z \leq C_0 s$ and $\alpha s z \leq (J_N^{\text{inv}})^2 s$. Adding and using $1 - \alpha + \alpha s \leq 1 + \alpha s$ proves it. The endpoint $\alpha = 0$ follows directly.

For the lower bound $T[1] = u$, and the unchanged polynomial 1 gives $\|[1]\|_{G_\alpha}^2 \leq m_k$. The $r = 1$ harmonic bound yields

$$\|T\|_{G_\alpha}^2 \geq \|u\|_{G_\alpha}^2/m_k \geq L_{1,N} h_N(\alpha)/m_k. \quad (\text{ACT11})$$

Consequently

$$\boxed{c_{-,N} = L_{1,N}/m_k, \quad c_{+,N} = C_0 + (J_N^{\text{inv}})^2, \quad c_{-,N} h_N(\alpha) \leq \|T\|_{G_\alpha}^2 \leq c_{+,N} h_N(\alpha).} \quad (\text{ACT12})$$

On the eventual original domain $s \geq 1$, $|\log c_{\pm,N}| = O_{h,A}(k \log^2(q+2))$. For c_+ , use the bounds on $D_N, F_N, b, J_N^{\text{inv}}$ and the fixed lower bound $J_N^{\text{inv}} \geq R_0^{-1}$; the latter follows because the joint sandwich requires $\mathbf{L}_k \mathbf{B}_k \geq 1$. For c_- , use $|\log L_{1,N}| \leq E_{1,N}$ and $|\log m_k| = O_h(k)$. The finite inequality holds before the eventual guard as well.

Order $\sigma_1(T) \geq \dots \geq \sigma_q(T) > 0$. On the codimension-at-most-one kernel of the rank-one row in (ACT8), T equals $B_{\alpha,N}$. Singular min-max therefore proves

$$\sigma_j(T; G_\alpha) \leq \overline{D}_N \quad (2 \leq j \leq q). \quad (\text{ACT13})$$

For the original M , order singular values increasingly. Then

$$s_1(M)^2 = \|T\|^{-2}, \quad s_j(M) \geq \overline{D}_N^{-1} \quad (j \geq 2), \quad -\log s_1(M)^2 = \log h_N(\alpha) + O_{h,A}(k \log^2(q+2)). \quad (\text{ACT14})$$

All remaining $q-1$ directions are retained.

If $L = \|T\| > \overline{D}_N$, the top singular value is simple by (ACT13). For unit vectors $Tv = Lw$, projecting (ACT8) away from $\mathbb{C}u$ gives $\|(I - P_u)w\| \leq \overline{D}_N/L$. The trace norm of the difference of two rank-one orthogonal projections is twice the sine of their angle. Hence

$$\|P_{\min,N} - P_u\|_1 \leq 2\overline{D}_N/L = 2\overline{D}_N s_1(M). \quad (\text{ACT15})$$

The top left singular line of T is exactly the minimum right singular line of M , with the original G_α -orthogonal projections.

4. Full observed heat and its actual observation fraction

The $r = 1$ harmonic enclosure gives

$$\theta_N(\alpha) = \frac{\|\Lambda u\|_{Q_\alpha}^2}{\|u\|_{G_\alpha}^2} \geq L_{1,N}/U_{1,N} > 0, \quad -\log \theta_N(\alpha) \leq 2E_{1,N}. \quad (\text{ACT16})$$

On the finite guard $\|T\|_{G_\alpha} > \overline{D}_N$ from (ACT15), the full positive heat $Z(\tau) = e^{-\tau M^\dagger M}$ satisfies, by (ACT13) and (ACT15),

$$\|Z(\tau) - e^{-\tau s_1^2} P_u\|_1 \leq 2\overline{D}_N s_1 e^{-\tau s_1^2} + (q-1)e^{-\tau/\overline{D}_N^2}. \quad (\text{ACT17})$$

The complete observed trace is

$$\Theta_N(\tau; \alpha) = \text{Tr}_B(\Lambda Z(\tau) L_\alpha) = \text{Tr}_E(P_{B,\alpha} Z(\tau)).$$

Since $P_{B,\alpha}$ is an orthogonal projection, $|\text{Tr}(P_{B,\alpha} X)| \leq \|X\|_1$, and $\text{Tr}(P_{B,\alpha} P_u) = \theta_N$. Therefore

$$\left| \frac{\Theta_N(\tau; \alpha)}{\theta_N e^{-\tau s_1^2}} - 1 \right| \leq \frac{2\overline{D}_N s_1}{\theta_N} + \frac{q-1}{\theta_N} e^{-\tau(\overline{D}_N^{-2} - s_1^2)}. \quad (\text{ACT18})$$

This is supplied (33), extending the original IFH11 relative estimate through the actual activation. The same finite guard is retained in (ACT18); (ACT12) and the following fixed-parameter critical scale prove it eventually.

Put $\ell_k = \log(q/k)$. The independently proved endpoints **HAR41-44** and harmonic clipping **HAR45-47** in **HARMONIC_PROOFS.md** give, in particular for fixed ζ , uniformly on bounded ζ -ranges,

$$\log h_N(\alpha_k(\zeta)) = 2q\ell_k + q\kappa_{L/H}(\zeta) + o(q), \quad \kappa_{L/H}(\zeta) = \min(\beta_{L/H}, \zeta). \quad (\text{ACT19})$$

This is the only additional limiting input used here. The scalar enclosure itself is uniform on the larger physical domain $\zeta \geq -2\ell_k$, as HAR47 proves; the singular separation and heat argument that follows uses the fixed or bounded parameter scope stated above. At $\tau_k(b) = e^{2q\ell_k + bq}$, the right side of (ACT18) tends to zero: $s_1 = e^{-q\ell_k + O(q)}$, $\overline{D}_N = e^{o(q)}$, and $\theta_N \geq e^{-o(q)}$. Moreover

$$\tau_k(b)s_1^2 = \exp\{q[b - \kappa_{L/H}(\zeta)] + o(q)\}.$$

When $\kappa_H(\zeta) < b < \kappa_L(\zeta)$, both low heat factors tend to one; both high factors are at most $e^{-e^{cq}}$ for some $c > 0$. The two low observation weights are at least $e^{-o(q)}$. Under the original signs $+, +, -, -$, they dominate the high terms, proving

$$\mathcal{R}\Theta_N(\tau_k(b); \alpha_k(\zeta)) > 0 \text{ eventually}, \quad q^{-1} \log \mathcal{R}\Theta_N(\tau_k(b); \alpha_k(\zeta)) \rightarrow 0. \quad (\text{ACT20})$$

No unweighted limit is assigned to this measured full heat.

For $\alpha = e^{-2\theta q}$, $\theta > 0$, the original scalar scales give $\alpha S_N \rightarrow \infty$, so the exact harmonic formula gives $\log h_N = 2\theta q + o(1)$. Thus $s_1 = e^{-\theta q + o(q)}$ while every s_j , $j \geq 2$, is at least $e^{-o(q)}$. The **full unobserved trace** at time $e^{2\sigma q}$ tends to one for $0 < \sigma < \theta$ and to zero for $\sigma > \theta$; the other $q - 1$ heat terms are bounded by $q \exp[-\exp(2\sigma q - o(q))] \rightarrow 0$.

5. The fixed observed arithmetic map and its full target minimum

Keep the actual nonzero conductor coefficients a_j , shifts

$$b_j = 4 + (2r_j - 8)\delta + i(2s_j - 8)\gamma, \quad \zeta_j = i(b_j - 4),$$

and $E_A(z) = \sum_j a_j e^{b_j z}$. Let $v = \text{ord}_0 E_A$, so $\mu_v \neq 0$ and $v \leq J - 1 \leq 80$. IVO1 gives $C_A f(y) = \sum_j a_j f(y - \zeta_j)$. Its symbol is $e^{-4z/i} E_A(z/i)$, whose first nonzero derivative is $i^{-v} \mu_v$. Expansion of the monomial therefore proves exactly

$$C_A[y^v] = i^{-v} \mu_v. \quad (\text{ACT21})$$

This needs no assumption $\Lambda[1] \neq 0$.

Define

$$W_r^{(v)} d = d_1[y^{-1}] + \sum_{j=2}^r d_j[y^{-(v+j)}].$$

It is the literal coordinate subspace with index set $J_v = \{1, v+2, \dots, v+r\}$ inside U_{r+v} ; (ACT2) makes $\Lambda W_r^{(v)}$ injective. Original multiplication gives

$$M^{v+1} W_r^{(v)} d = d_1[y^v] + \sum_{j=2}^r d_j[y^{-(j-1)}]. \quad (\text{ACT22})$$

We prove the distance of the first target column from the others through every original observed-fibre correction. Put

$$L = J(r-1), \quad B(y) = \prod_j (y - \zeta_j)^{r-1},$$

$$t_* = \max(1, (R_- + Z)/(2\delta)), \quad R_- = (k-8)R_0, \quad Z = \max_j |\zeta_j|.$$

For every $d \in \mathbb{C}^{r-1}$,

$$f(y) = C_A(y^v + U_{r-1}d) = i^{-v}\mu_v + \sum_{j,n} \frac{a_j d_n}{(y - \zeta_j)^n}.$$

The polynomial Bf has degree L and leading coefficient $i^{-v}\mu_v$; each proper rational term has numerator degree at most $L - 1$.

Choose $L + 1$ distinct original lower roots $\eta_0, \dots, \eta_L$. The guard $L + 1 \leq q_-$ permits this. Distances are at least 2δ , moduli at most R_- , and none is a pole. Lagrange's exact leading-coefficient formula gives

$$i^{-v}\mu_v = \sum_{s=0}^L \frac{B(\eta_s)f(\eta_s)}{\prod_{t \neq s} (\eta_s - \eta_t)}.$$

The coefficients have moduli at most t_*^L . Cauchy-Schwarz proves

$$\|C_A(y^v + U_{r-1}d)\|_2^2 \geq \frac{|\mu_v|^2}{(L+1)t_*^{2L}}. \quad (\text{ACT23})$$

The full lower-root norm is at least its selected-node norm.

Each shifted lower grid is a subset of the upper grid, so each restriction has Euclidean norm at most one. Thus

$$\|C_A w\|_2 \leq A_1 \|w\|_2 \leq A_1 a_{\min}^{-k/2} \|w\|_{T_k}, \quad A_1 = \sum_j |a_j|.$$

IVO2 gives $\ker \Lambda \subseteq \ker C_A$. Every lift w of the observed class $\Lambda(y^v + U_{r-1}d)$ therefore has the same conductor image. Since $G_\alpha \succeq \mathbf{L}_k^{-1} T_k$,

$$\|w\|_{G_\alpha}^2 \geq \frac{a_{\min}^k}{\mathbf{L}_k A_1^2} \|C_A(y^v + U_{r-1}d)\|_2^2.$$

Minimizing first over the entire original observed fibre and then over d gives

$$d_- \leq \rho := \text{dist}_{Q_\alpha}(\Lambda[y^v], \Lambda U_{r-1})^2 \leq d_+, \quad (\text{ACT24})$$

$$d_- = \frac{a_{\min}^k |\mu_v|^2}{\mathbf{L}_k A_1^2 (L+1) t_*^{2L}}, \quad d_+ = u_k \int_{\mathbb{R}} y^{2v} d\sigma(y).$$

The upper bound uses $d = 0$, the allowed polynomial y^v , $G_\alpha \preceq G_0$, and the original source comparison. In fact it also proves

$$a := \|\Lambda[y^v]\|_{Q_\alpha}^2 \leq d_+. \quad (\text{ACT25})$$

This bound on the **whole first-column norm**, not only ρ , controls the shear below.

The original Gamma moment is finite and positive. With fixed original nonzero μ_v , the inherited source costs give

$$\max(|\log d_-|, |\log d_+|) = O_{h,A}(k \log^2(q+2) + Jr \log(q+2)) = o(q). \quad (\text{ACT26})$$

This concerns the fixed admitted period stratum and its actual first nonzero moment; it does not assert a uniform constant while that moment tends arbitrarily fast to zero.

Equation (ACT24) proves independence of $\Lambda[y^v], \Lambda U_{r-1}$. Therefore the map between the exact original observed subspaces

$$\mathcal{A}_r^{(v)}(\Lambda W_r^{(v)} d) = \Lambda M^{v+1} W_r^{(v)} d \quad (\text{ACT27})$$

is well defined and bijective onto its target. Both metrics are restrictions of Q_α . This is not $(\Lambda M L_\alpha)^{v+1}$.

6. Exact common-subspace cancellation in the determinant

Let $I = \{1, \dots, r-1\}$, $J = J_v$, $C = I \cap J$, and denote the corresponding original observed inverse-power Grams by H_I, H_J, H_C . They come from the same $H_{r+v,\alpha}^B$. Schur factorization gives

$$\det H_I = \det H_C \det S_I, \quad \det H_J = \det H_C \det S_J,$$

with empty determinant one. These are complete attained coefficient metrics after minimizing over the same common coordinate subspace.

Put $h = h_N(\alpha)$, $E = E_{r+v,N}$, $l = e^{-E}$, $u_* = e^E$. The harmonic enclosure gives

$$lhI \preceq S_I, S_J \preceq u_*hI. \quad (\text{ACT28})$$

For the lower inequality, every extension of a specified remaining coefficient vector has at least its Euclidean norm; for the upper inequality, set eliminated coordinates to zero. These two facts and (ACT2), followed by the actual minima, prove the claim without interchanging restriction and inversion.

Write $i = |I \setminus C|$, $j = |J \setminus C|$. Then $j = i+1$ and $i+j \leq 2v+1$: for $r \geq v+3$, $i = v, j = v+1$; for $2 \leq r \leq v+2$, $C = \{1\}$ and $i+j = 2r-3$; for $r = 1$, $i = 0, j = 1$.

The target determinant is $\rho \det H_I$, while the source determinant is $\det H_J$. Hence exactly

$$\mathcal{J}_{r,N}^{(v)}(\alpha) = \rho \det S_I / \det S_J.$$

Therefore

$$|\log \mathcal{J}_{r,N}^{(v)}(\alpha) + \log h_N(\alpha)| \leq (2v+1)E_{r+v,N} + \max(|\log d_-|, |\log d_+|). \quad (\text{ACT29})$$

The common determinant cancels before estimation. There is no factor r multiplying its error.

The right side is $o(q)$. From (ACT19), with original four-cutoff signs,

$$\mathcal{R} \log \mathcal{J}_{r,N}^{(v)}(\alpha_k(\zeta)) = -2q[\kappa_L(\zeta) - \kappa_H(\zeta)] + o(q) = -\mathfrak{C}_{\text{inv}}(\zeta)q + o(q). \quad (\text{ACT30})$$

At the canonical source the result is $-\mathfrak{C}q + o(q)$. This proves supplied (39)–(40) and (3).

7. Every action singular value, with the complete shear

The determinant does not by itself determine all singular scales. The extra estimate is (ACT25).

In the literal source coefficient basis, its Gram is $S = H_J$; write the target Gram

$$V = \begin{pmatrix} a & b^* \\ b & K \end{pmatrix}, \quad K = H_{r-1,\alpha}^B.$$

For $r = 1$, interpret K, b as empty. Put

$$c = K^{-1}b, \quad \rho = a - b^*K^{-1}b, \quad \mathcal{T}_c = \begin{pmatrix} 1 & 0 \\ c & I \end{pmatrix}.$$

The complete square is the exact congruence

$$V = \mathcal{T}_c^* \begin{pmatrix} \rho & 0 \\ 0 & K \end{pmatrix} \mathcal{T}_c. \quad (\text{ACT31})$$

All cross terms remain. From $K \succeq lhI$ and $a \leq d_+$,

$$\|c\|^2 = b^* K^{-2} b \leq \frac{b^* K^{-1} b}{lh} \leq \frac{d_+}{lh}. \quad (\text{ACT32})$$

The nontrivial singular values of $\mathcal{T}_c$ are

$$\chi(c) = \frac{\sqrt{4 + \|c\|^2} + \|c\|}{2}, \quad \chi(c)^{-1}.$$

Its restriction to the span of the first coordinate and $(0, c)$ has squared-singular determinant one and trace $2 + \|c\|^2$; on the orthogonal complement it is the identity. This proves the formula and $\|\mathcal{T}_c\| = \|\mathcal{T}_c^{-1}\| = \chi(c)$.

Let $\delta_1 \leq \dots \leq \delta_r$ be the eigenvalues of $\text{diag}(\rho, K)$, and $s_1 \leq \dots \leq s_r$ all the singular values of (ACT27) in its original metrics. Their squares are the generalized eigenvalues of (V, S) . Congruence (ACT31), min-max, and $lhI \preceq S \preceq u_* hI$ give

$$\boxed{\frac{\delta_j}{u_* h \chi(c)^2} \leq s_j^2 \leq \frac{\chi(c)^2 \delta_j}{lh}, \quad 1 \leq j \leq r.} \quad (\text{ACT33})$$

For the congruence estimate, substitute $z = \mathcal{T}_c x$ in the Rayleigh quotient and use $\chi^{-2} \|z\|^2 \leq \|x\|^2 \leq \chi^2 \|z\|^2$. The invertible substitution preserves dimensions and sends all subspaces bijectively to subspaces in the min-max formula. The source comparison is a second Rayleigh estimate on those same coefficient subspaces.

Equation (ACT33) holds at every activation on the finite domain, recording all shear amplification. On the explicit finite guard $lh \geq d_+$, the smallest diagonal eigenvalue is ρ , and the other $r - 1$ are the eigenvalues of K . Thus

$$\boxed{\frac{d_-}{u_* h \chi^2} \leq s_1^2 \leq \frac{d_+ \chi^2}{lh}, \quad \frac{l}{u_* \chi^2} \leq s_j^2 \leq \frac{u_* \chi^2}{l} \quad (2 \leq j \leq r).} \quad (\text{ACT34})$$

The upper bound on s_1^2 improves to $d_+/(lh)$ by testing $(1, -c)$, whose target squared norm is ρ .

At critical activation, (ACT19), $E = o(q)$, and (ACT26) imply

$$\log(lh/d_+) = 2q\ell_k + O(q) \rightarrow +\infty.$$

Hence the finite guard holds eventually, and $\|c\| \leq e^{-q\ell_k + O(q)} \rightarrow 0$, so $\chi(c) \rightarrow 1$. Every singular direction therefore satisfies

$$-\log s_1^2 = \log h_N(\alpha_k(\zeta)) + o(q), \quad |\log s_j^2| = o(q) \quad (2 \leq j \leq r), \quad (\text{ACT35})$$

uniformly in j . The smallest value is eventually simple by its strict separation from the lower bound for s_2 .

The complete unweighted domain heat obeys

$$e^{-\tau s_1^2} \leq \text{Tr}_{\Lambda W_r^{(v)}} e^{-\tau(\mathcal{A}_r^{(v)})^\dagger \mathcal{A}_r^{(v)}} \leq e^{-\tau s_1^2} + (r - 1)e^{-\tau l/(u_* \chi^2)}. \quad (\text{ACT36})$$

At $\tau_k(b) = e^{2q\ell_k + bq}$, its second term tends to zero: $r \leq q$, $l/(u_* \chi^2) = e^{-o(q)}$. The first term tends to one at both low cutoffs and zero at both high cutoffs whenever $\kappa_H(\zeta) < b < \kappa_L(\zeta)$. Therefore

$$\boxed{\mathcal{R} \text{Tr} e^{-\tau_k(b)(\mathcal{A}_r^{(v)})^\dagger \mathcal{A}_r^{(v)}} \rightarrow 2.} \quad (\text{ACT37})$$

This proves the unweighted action heat through every singular value.

The need for control of a has an exact two-dimensional example. Take $h > 1$, source $S = hI_2$, and target

$$V_h = \begin{pmatrix} 1 + h^3 & h^2 \\ h^2 & h \end{pmatrix}.$$

Here $K = h, \rho = 1, \det V_h / \det S = 1/h$, but the shear is $c = h$. The squared singular values have product $1/h$ and sum $h^2 + 1 + 1/h$; one grows like h^2 , the other decays like h^{-3} . A controlled inverse block and Schur remainder alone would not prove (ACT35). The original bound $a \leq d_+$ excludes precisely this uncontrolled amplification.

8. The original small-regularizer edge and full source cost

Use the fixed original G_0 -isometric frame

$$M_0 = G_0^{1/2} M G_0^{-1/2}, \quad C_\alpha = G_0^{1/2} G_\alpha^{-1} G_0^{1/2} = I + \alpha K_N.$$

The metric there is C_α^{-1} and $M_0^\dagger = C_\alpha M_0^* C_\alpha^{-1}$. The identity $\det(aI + XY) = \det(aI + YX)$ gives

$$\det(aI + M^\dagger M) = \frac{\det(aC_\alpha + M_0 C_\alpha M_0^*)}{\det C_\alpha} =: \frac{\mathcal{P}_N(a, \alpha)}{\det(I + \alpha K_N)}. \quad (\text{ACT38})$$

This rederives dual pencil D2 at the exact original activation. At zero,

$$\mathcal{P}_N(0, \alpha) = |\det M|^2 \det C_\alpha, \quad \Delta_N(\alpha) = \log \det G_0 - \log \det G_\alpha = \log \det C_\alpha. \quad (\text{ACT39})$$

For $T = M^{-1}$, $(M^\dagger M)^{-1} = TT^\dagger$, and $T^\dagger T$ has the same eigenvalues. Factoring $M^\dagger M$ and applying the determinant identity proves

$$\boxed{\mathcal{P}_N(a, \alpha) / \mathcal{P}_N(0, \alpha) = \det(I + aT^\dagger T)}. \quad (\text{ACT40})$$

Adjoint use G_α . This is a determinant equality, not an operator equality.

All q singular directions are in

$$\log \det(I + aT^\dagger T) = \log(1 + a\|T\|^2) + \sum_{j=2}^q \log(1 + a\sigma_j(T)^2).$$

By (ACT13) and $0 \leq \log(1 + x) \leq x$,

$$0 \leq \log \det(I + aT^\dagger T) - \log(1 + a\|T\|^2) \leq (q-1)a\bar{D}_N^2. \quad (\text{ACT41})$$

At $a_k(b) = e^{-2q\ell_k - bq}$, this upper remainder tends to zero. Equations (ACT12) and (ACT19) yield $\log(a_k(b)\|T\|^2) = q[\kappa_{L/H}(\zeta) - b] + o(q)$. Since $0 \leq \log(1 + e^x) - x_+ \leq \log 2$,

$$\boxed{\frac{1}{q} \log \frac{\mathcal{P}_N(a_k(b), \alpha_k(\zeta))}{\mathcal{P}_N(0, \alpha_k(\zeta))} \rightarrow [\kappa_{L/H}(\zeta) - b]_+}. \quad (\text{ACT42})$$

Threshold equality is included.

The exact two-metric receiver retains the full source cost:

$$\log \frac{\mathcal{P}_N(a, \alpha)}{\mathcal{P}_N(a, 0)} - \Delta_N(\alpha) = \log \det(I + aT^\dagger G_\alpha T) - \log \det(I + aT^\dagger G_0 T). \quad (\text{ACT43})$$

Indeed division of (ACT40) at α by its value at zero leaves $\mathcal{P}_N(0, \alpha) / \mathcal{P}_N(0, 0) = \det C_\alpha$, with logarithm exactly Δ_N .

For general critical ζ, b , the four-cutoff return in (ACT43), divided by q , therefore tends to

$$2\{[\min(\beta_L, \zeta) - b]_+ - [\min(\beta_H, \zeta) - b]_+ - [\beta_L - b]_+ + [\beta_H - b]_+\}. \quad (\text{ACT44})$$

This gives the complete clipped small-regularizer receiver, including thresholds.

9. Concrete simultaneous activation, action volume, and heat

Put

$$a_0 = \log(4/\pi) > 0, \quad \beta_* = -2 - I_\square, \quad \beta_L = \beta_* + 2a_0, \quad \beta_H = \beta_* + 2a_0 - \mathfrak{C}/2, \quad \zeta_\diamond = \beta_* + a_0.$$

The inherited certified interval $1353893/10^6 < \mathfrak{C} < 1354751/10^6$ is S9 in the pinned source, with $\mathfrak{C} = -\mathfrak{c}_{\min}$. It implies $\mathfrak{C} > 4a_0$: $\pi > 3$ and $\log(4/3) < 1/3$ give $4a_0 < 4/3 < 1353893/10^6$. Consequently

$$\beta_H < \beta_* < \zeta_\diamond < \beta_L. \quad (\text{ACT45})$$

Take the original explicit values

$$\alpha_k^\diamond = e^{-2q\ell_k - \zeta_\diamond q}, \quad \tau_k^* = e^{2q\ell_k + \beta_* q}.$$

The latter is exactly the earlier original late time IFH13.

Since $\kappa_L(\zeta_\diamond) = \zeta_\diamond$ and $\kappa_H(\zeta_\diamond) = \beta_H$, $2(\kappa_L - \kappa_H) = \mathfrak{C} - 2a_0$. For $r = k$, the shared harmonic determinant calculation and the action proof give

$$\mathcal{R} \log \det H_{k,N}^B(\alpha_k^\diamond) = (\mathfrak{C} - 2a_0)kq + o_{h,A}(kq), \quad (\text{ACT46})$$

$$\mathcal{R} \log \mathcal{J}_{k,N}^{(v)}(\alpha_k^\diamond) = -(\mathfrak{C} - 2a_0)q + o_{h,A}(q), \quad \mathcal{R} \text{Tr} e^{-\tau_k^* (\mathcal{A}_k^{(v)})^\dagger \mathcal{A}_k^{(v)}} \rightarrow 2. \quad (\text{ACT47})$$

Equation (ACT46) uses the shared harmonic determinant conclusion; both conclusions in (ACT47) are derived here from the fixed original arithmetic map.

Finally set $a_k = 1/\tau_k^*$, hence $b = \beta_*$ in (ACT44). The low positive parts are a_0 after activation and $2a_0$ canonically; both high positive parts vanish by (ACT45). Therefore

$$\mathcal{R} \left[\log \frac{\mathcal{P}_N(a_k, \alpha_k^\diamond)}{\mathcal{P}_N(a_k, 0)} - \Delta_N(\alpha_k^\diamond) \right] = -2 \log(4/\pi) q + o_{h,A}(q). \quad (\text{ACT48})$$

This is supplied (48), with the complete original source cost and all inverse singular directions.

10. Verification scope

check_action.py supplies exact rational fixtures for the polynomial-section inverse, covariance coisometry and moving dual row, conductor phase and actual action columns, observed metrics with nonzero kernel corrections, common-subspace determinant cancellation, shear congruence, and spectral-polynomial edge identity. It also tests the exact two-dimensional amplification example and finite-precision enclosures. Small finite grids are algebraic fixtures, not claims that their asymptotics meet the large- k guards. The executed receipt is **ACTION_CHECK_RESULTS.json**.

Part IV

Complete received critical-activation continuation

The new inverse-power estimate is strong enough to determine the critical source-activation layer, including the original observation. It also yields an evaluated determinant for an actual arithmetic action on a growing observed subspace.

The main coefficient is a clipped affine function. Define

$$a_0 = \log(4/\pi),$$

$$I_{\square} = \log(\delta^2 + \gamma^2) - 3 + \frac{\delta}{\gamma} \arctan \frac{\gamma}{\delta} + \frac{\gamma}{\delta} \arctan \frac{\delta}{\gamma},$$

and retain the incoming constant

$$\mathfrak{C} = 4[\psi(0) - \psi(1) - J(1) - 1].$$

Put

$$\beta_L = 2a_0 - 2 - I_{\square}, \quad \beta_H = \beta_L - \frac{\mathfrak{C}}{2}.$$

At the original activation

$$\alpha_k(\zeta) = \exp[-2q \log(q/k) - \zeta q],$$

the complete observed inverse-power determinant satisfies

$$\mathcal{R} \log \det H_{r,N}^B(\alpha_k(\zeta)) = rq \mathfrak{C}_{\text{inv}}(\zeta) + o_{h,A}(rq), \quad (1)$$

where

$$\mathfrak{C}_{\text{inv}}(\zeta) = 2[\min(\beta_L, \zeta) - \min(\beta_H, \zeta)] = \begin{cases} 0, & \zeta \leq \beta_H, \\ 2(\zeta - \beta_H), & \beta_H < \zeta < \beta_L, \\ \mathfrak{C}, & \zeta \geq \beta_L. \end{cases} \quad (2)$$

This holds for every r with $r \log(q+2) = o(q)$, in particular $r = k$. It evaluates the whole transition at the requested kq scale.

There is a second result in the original arithmetic action, rather than just its metric. Let $v \leq 80$ be the first nonzero conductor-moment order. On the explicitly specified observed space generated by

$$[y^{-1}], [y^{-(v+2)}], \dots, [y^{-(v+r)}],$$

apply the original M^{v+1} , then the original observation. Its full squared exterior-volume ratio, denoted $\mathcal{J}_{r,N}^{(v)}(\alpha)$, satisfies

$$\mathcal{R} \log \mathcal{J}_{r,N}^{(v)}(\alpha_k(\zeta)) = -\mathfrak{C}_{\text{inv}}(\zeta) q + o_{h,A}(q). \quad (3)$$

Every one of its r singular directions is included.

At a concrete activation and the **same late heat time used in the new attachment**, this observed arithmetic map has the unweighted heat return

$$\boxed{\mathcal{R} \operatorname{Tr} \exp \left[-\tau_k^* (\mathcal{A}_r^{(v)})^\dagger \mathcal{A}_r^{(v)} \right] \longrightarrow 2.} \quad (4)$$

The map $\mathcal{A}_r^{(v)}$, its domain, its target, and the adjoint metric are specified below. It is not a power of an independently compressed operator.

The finite comparison behind these results uses the supplied polynomial estimates and the complete joint-source minimum. The limiting constants additionally use the incoming Gamma monic-profile and zero-law inputs, at their stated proof status. The incoming theorem provides the canonical observed rq -return and the explicit finite bounds; the activation layer and arithmetic-action calculation below are the continuation.

1. Keep the original inverse powers, observation, and source activation

Work on the attachment's simple-quartet domain

$$q = (k+1)^2, \quad k \equiv 1 \pmod{4}, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2.$$

Write

$$R_0 = \sqrt{\delta^2 + \gamma^2}, \quad R = kR_0,$$

$$Q(y) = \prod_{a,b=0}^k [y - (2b - k)\gamma + i(2a - k)\delta], \quad E_k = \mathbb{C}[y]/(Q),$$

and

$$M[P] = [yP].$$

The physical coordinate remains $S = k/2 + iy$, the source remains $w_h^{*k}(y) dy$, and the full arithmetic unit remains in the original tensor-value map.

The original scalar and joint minima are G_N and G_N^J . Retain their exact source activation:

$$\boxed{G_N(\alpha)^{-1} = (1 - \alpha)G_N^{-1} + \alpha(G_N^J)^{-1}, \quad 0 \leq \alpha \leq 1.} \quad (5)$$

At intermediate α , the normal-source penalty is the specified one; it is not represented as the unchanged physical norm. The preceding proof supplies

$$\mathbf{L}_k^{-1} T_k \preceq G_N^J \preceq \mathbf{B}_k T_k, \quad \log(\mathbf{L}_k \mathbf{B}_k) = O_h(k \log^2(q+2)), \quad (6)$$

on the same complete value space. In root-value coordinates,

$$T_k = \operatorname{diag}(W_\omega), \quad W_\omega \geq a_{\min}^k, \quad \sum_{\omega} W_\omega = A_w^k.$$

These weights count the original ordered tuples.

For the original observation $\Lambda : E_k \rightarrow B_k$, define

$$Q_{B,N}(\alpha) = (\Lambda G_N(\alpha)^{-1} \Lambda^*)^{-1}.$$

For the literal inverse-power frame

$$U_r d = \sum_{j=1}^r d_j [y^{-j}],$$

put

$$H_{r,N}^E(\alpha) = U_r^* G_N(\alpha) U_r,$$

$$H_{r,N}^B(\alpha) = U_r^* \Lambda^* Q_{B,N}(\alpha) \Lambda U_r. \quad (7)$$

The observation, period, and kernel are fixed. Only the specified metric changes. Let

$$n_0 = 2 \left\lfloor \frac{N+1-q}{2} \right\rfloor, \quad K_N = \mathsf{K}_{Q,n_0}(0,0),$$

where the kernel is for the original Gamma relation measure $Q^2 d\sigma$. Introduce the scalar

$$\boxed{S_N = \frac{u_k}{|Q(0)|^2 K_N}}. \quad (8)$$

Here u_k is the same original source-comparison constant at every cutoff. It is not used to divide or redefine the arithmetic measure.

The incoming finite proof gives

$$a_{r,N} S_N I_r \preceq H_{r,N}^B(0) \preceq H_{r,N}^E(0) \preceq b_{r,N} S_N I_r, \quad (9)$$

with explicitly specified positive constants and

$$\log^+(a_{r,N}^{-1}) + \log^+ b_{r,N} = O_{h,A}((k+Jr) \log(q+Jr+2)).$$

For example, in the attachment's finite notation one may take

$$a_{r,N} = \frac{\ell_{k,N}}{u_k} \left[\frac{\mathcal{LR}_*}{2\mathcal{F}_N \mathcal{B}^{Jr}} \right]^2, \quad b_{r,N} = \mathcal{U}_{r,N}^2.$$

These contain its actual pole coefficients, root products, and complete Gamma minima.

The obstacle to transporting (9) through (5) is that restriction and inversion do not commute. The next calculation controls that obstacle on the original full space.

2. The full covariance cost is bounded by the same inverse-power scale

Let

$$\omega_{ab} = (2b-k)\gamma - i(2a-k)\delta, \quad D_{ab} = |Q'(\omega_{ab})|.$$

For $b < k$, cancellation of the complete adjacent products gives

$$\boxed{\frac{D_{a,b+1}^2}{D_{ab}^2} = \prod_{m=0}^k \frac{\delta^2(a-m)^2 + \gamma^2(b+1)^2}{\delta^2(a-m)^2 + \gamma^2(k-b)^2}}. \quad (10)$$

The omitted self-factor is accounted for in the row $m = a$.

Every factor in this ratio is below one before the two central indices, equal to one between them, and above one afterwards. The same calculation in the other index proves that the four central roots minimize $|Q'|$ **exactly**.

Choose one of them, ω_c , so that $|\omega_c| = R_0$. A lattice shell of maximum displacement j contains at most $8j$ roots, each at distance at least $2\delta j$ from ω_c . Hence

$$\begin{aligned} \log \frac{|Q(0)|}{R_0|Q'(\omega_c)|} &= \sum_{\omega \neq \omega_c} \log \frac{|\omega|}{|\omega - \omega_c|} \\ &\leq R_0 \sum_{\omega \neq \omega_c} \frac{1}{|\omega - \omega_c|} \leq \frac{4kR_0}{\delta}. \end{aligned}$$

Therefore

$$\boxed{\frac{|Q(0)|}{|Q'(\omega)|} \leq R_0 e^{4kR_0/\delta} \quad \text{for every original root } \omega.} \quad (11)$$

This finite inequality is stronger for the present purpose than a generic Vandermonde condition-number estimate.

Now use the attachment's exact normalized kernel polynomial

$$H(y) = \frac{K_{Q,n_0}(y, 0)}{K_N}.$$

Its proved hard-gap estimate gives

$$|H(\omega)| \geq \exp[-n_0 R^2 / \beta_q], \quad \beta_q = 2^{-27} q^2,$$

on the stated eventual guards.

The polynomial

$$P_\omega(y) = \frac{Q(y)H(y)}{(y - \omega)Q'(\omega)H(\omega)}$$

has degree at most N and represents the original cardinal class e_ω . Since

$$|y - \omega| \geq \delta \quad (y \in \mathbb{R}),$$

its source norm gives

$$\|e_\omega\|_{G_N}^2 \leq \frac{u_k e^{2n_0 R^2 / \beta_q}}{\delta^2 |Q'(\omega)|^2 K_N}.$$

Consequently,

$$\boxed{\text{Tr}(T_k^{-1} G_N) \leq \Gamma_N S_N,} \quad (12)$$

where one explicit choice is

$$\boxed{\Gamma_N = \max \left\{ 1, \frac{qR_0^2}{\delta^2 a_{\min}^k} \exp \left(\frac{8kR_0}{\delta} + \frac{2n_0 R^2}{\beta_q} \right) \right\}.} \quad (13)$$

Its logarithm is $O_h(k + \log q)$.

Thus the **whole canonical covariance comparison** costs only a sub- q logarithmic factor relative to S_N . This is what makes a sharp critical-layer calculation possible.

3. An all-activation harmonic bound, with the full observation minimum retained

Set

$$F = \Lambda U_r, \quad H_0 = F^* Q_{B,N}(0) F,$$

and use its actual canonical left inverse

$$R_N = H_0^{-1} F^* Q_{B,N}(0).$$

Then

$$R_N F = I_r, \quad R_N Q_{B,N}(0)^{-1} R_N^* = H_0^{-1}. \quad (14)$$

For any positive observed metric Q_B , its complete minimum under the value map R_N gives

$$F^* Q_B F \succeq (R_N Q_B^{-1} R_N^*)^{-1}.$$

This does not replace the inverse of the full covariance by an inverse on a restricted subspace.

The elementary matrix inequality

$$T_k^{-1} \preceq \text{Tr}(T_k^{-1} G_N) G_N^{-1}$$

and (12) imply

$$H_{r,N}^B(\alpha) \succeq \frac{H_0}{1 - \alpha + \alpha \mathbf{L}_k \Gamma_N \mathbf{S}_N}. \quad (15)$$

For the upper bound, both original minima remain available:

$$H_{r,N}^E(\alpha) \preceq \frac{H_{r,N}^E(0)}{1 - \alpha},$$

$$H_{r,N}^E(\alpha) \preceq \frac{\mathbf{B}_k}{\alpha} U_r^* T_k U_r.$$

Because every original root has modulus at least $R_0 > 2$,

$$U_r^* T_k U_r \preceq A_w^k \sum_{j=1}^r R_0^{-2j} I_r \preceq \frac{A_w^k}{R_0^2 - 1} I_r.$$

Define

$$h_N(\alpha) = \frac{\mathbf{S}_N}{1 - \alpha + \alpha \mathbf{S}_N}. \quad (16)$$

Equations (9), (15), and the two upper bounds prove

$$L_{r,N} h_N(\alpha) I_r \preceq H_{r,N}^B(\alpha) \preceq H_{r,N}^E(\alpha) \preceq U_{r,N} h_N(\alpha) I_r, \quad (17)$$

where

$$L_{r,N} = \frac{a_{r,N}}{\max(1, \mathbf{L}_k \Gamma_N)}, \quad U_{r,N} = b_{r,N} + \frac{\mathbf{B}_k A_w^k}{R_0^2 - 1}.$$

For the upper bound, the only scalar step is

$$\min\left(\frac{bs}{1-\alpha}, \frac{d}{\alpha}\right) \leq (b+d)\frac{s}{1-\alpha+\alpha s},$$

with the continuous endpoint interpretation.

Put

$$E_{r,N} = \max\{0, -\log L_{r,N}, \log U_{r,N}\}.$$

Then

$$E_{r,N} = O_{h,A}\left(k \log^2(q+2) + (k+Jr) \log(q+Jr+2)\right) = o(q). \quad (18)$$

The last equality uses $r \log(q+2) = o(q)$.

This proves a stronger observation theorem than the canonical one:

$$e^{-2E_{r,N}} H_{r,N}^E(\alpha) \preceq H_{r,N}^B(\alpha) \preceq H_{r,N}^E(\alpha) \quad \text{for every } 0 \leq \alpha \leq 1. \quad (19)$$

The entire original observation kernel and its source mixing are present.

For $X = E$ or B ,

$$|\log \det H_{r,N}^X(\alpha) - r \log h_N(\alpha)| \leq r E_{r,N}. \quad (20)$$

At the four cutoffs, n_0 is exactly $0, 0, q, q$, so there are exactly two scalar values S_L, S_H . Thus

$$\begin{aligned} \mathcal{R} \log \det H_{r,N}^X(\alpha) &= 2r \log \frac{S_L(1-\alpha+\alpha S_H)}{S_H(1-\alpha+\alpha S_L)} + \varepsilon_{r,k}, \\ |\varepsilon_{r,k}| &\leq r \sum_{\text{four } N} E_{r,N}. \end{aligned} \quad (21)$$

Equation (21) is a finite scalar enclosure for the complete return at every activation. It is not merely a formula involving another unresolved arithmetic Gram.

4. Evaluate the two scalar endpoints and the critical layer

The complete lattice product gives

$$\log |Q(0)|^2 = 2q \log k + q I_{\square} + O_h(k \log(k+2)). \quad (22)$$

To control the logarithmic singularity, isolate the central cells and sum the variation over the remaining lattice rings. Their total error is $O_h(k \log k)$; no central root is discarded.

The fixed rectangular integral is

$$I_{\square} = \frac{1}{4\delta\gamma} \int_{-\delta}^{\delta} \int_{-\gamma}^{\gamma} \log(x^2 + y^2) dy dx,$$

and direct integration gives the formula stated at the beginning.

The Gamma product yields the global bounds

$$\frac{\sigma_0 e^{-\pi|y|/2}}{1+4y^2} \leq \sigma(y) \leq \sigma_0 e^{1/2} (1+4y^2)^{1/4} e^{-\pi|y|/2},$$

where $\sigma_0 = \Gamma(1/4)^2/(2\pi)$. Comparing the full root polynomial with $(|y|+R)^q$ above and $(|y|-R)^q$ on $|y| \geq R+1$ below gives

$$\log \nu_0^\Gamma = 2q \log q + (2a_0 - 2)q + O_h(k + \log q).$$

The source comparison at the constant polynomial gives

$$|\log u_k| = O_h(k + \log q).$$

Therefore

$$\boxed{\log S_L = 2q\ell_k + \beta_L q + o_h(q)}. \quad (23)$$

The incoming exact kernel identity, followed by its stated monic-profile and reference-zero-law calculation, gives

$$\log \frac{\mathcal{K}_{Q,q}(0,0)}{\mathcal{K}_{Q,0}(0,0)} = \frac{\mathfrak{C}}{2}q + o_h(q).$$

Hence

$$\boxed{\log S_H = 2q\ell_k + \beta_H q + o_h(q)}. \quad (24)$$

This is the precise place where the supplied analytic coefficient enters. Its interval

$$1.353893 < \mathfrak{C} < 1.354751$$

is inherited from that proof, not a new numerical calculation here.

When $S_N \geq 1$,

$$|\log h_N(\alpha) - \min(\log S_N, -\log \alpha)| \leq \log 2.$$

Consequently, for

$$\alpha_k(\zeta) = e^{-2q\ell_k - \zeta q},$$

every eigenvalue of either inverse-power Gram satisfies

$$\boxed{\log \lambda_j(H_{r,N}^X(\alpha_k(\zeta))) = 2q\ell_k + q \min(\beta_{L/H}, \zeta) + o_{h,A}(q)}. \quad (25)$$

The error is uniform over all r directions and ζ in compact sets. Equations (20)–(25) prove (1)–(2).

The coarser activation $e^{-2bq\ell_k}$ consequently has two different inverse-power regimes:

$$\frac{1}{rq} \mathcal{R} \log \det H_{r,N}^X(e^{-2bq\ell_k}) \longrightarrow \begin{cases} 0, & 0 \leq b < 1, \\ \mathfrak{C}, & b \geq 1. \end{cases} \quad (26)$$

The endpoint $b = 1$ is included because

$$I_\square > 2 \log \gamma - 2 \implies \beta_L < 2 \log \frac{4}{\pi \gamma} < 0.$$

The finer ζ -layer resolves the transition hidden by (26).

5. The original arithmetic inverse has one small singular direction at this scale

This can be derived directly from the original polynomial source.

Let R_N now denote the canonical scalar minimum section, and define

$$\mathcal{D}p = \frac{p - p(0)}{y}, \quad B_N = J_N \mathcal{D}R_N, \quad \ell_N(x) = (R_N x)(0), \quad u = [y^{-1}].$$

Then

$$\boxed{M^{-1} = B_N + u \ell_N.} \quad (27)$$

The attachment's vector $-Q(0)[y^{-1}]$ spans the same line; its factor is retained in S_N and in the matched dual row.

The original source comparison and anchored-division estimate give

$$\|B_N\|_{G_N} \leq \mathfrak{d}_N \sqrt{u_k / \ell_{k,N}} =: D_N.$$

Also,

$$\|\ell_N\|_{G_N^{-1}} \leq \sqrt{\frac{N+1}{\ell_{k,N} \sqrt{2\pi}}},$$

because the Gamma evaluation kernel at zero is at most $(N+1)/\sqrt{2\pi}$. These are the same finite polynomial estimates used in the incoming proof.

In the joint metric,

$$\|M^{-1}\|_{G_N^J} \leq \frac{\sqrt{\mathbf{L}_k \mathbf{B}_k}}{R_0}.$$

Combining the covariance coisometry with these bounds gives finite positive constants $c_{-,N}, c_{+,N}$, with

$$|\log c_{\pm,N}| = O_{h,A}(k \log^2(q+2)),$$

such that

$$\boxed{c_{-,N} h_N(\alpha) \leq \|M^{-1}\|_{G_N(\alpha)}^2 \leq c_{+,N} h_N(\alpha).} \quad (28)$$

For the lower bound, simply apply M^{-1} to the unchanged constant class:

$$M^{-1}[1] = u, \quad \|[1]\|_{G_N(\alpha)}^2 \leq M_h(0)^k.$$

For the upper bound, combine the canonical bound from (27) with

$$G_N^J \preceq G_N(\alpha) \preceq \alpha^{-1} G_N^J.$$

This gives matching multiples of $\min(S_N, \alpha^{-1})$, which differs from $h_N(\alpha)$ by at most a factor two.

Therefore

$$\boxed{-\log s_{\min}(M; G_N(\alpha))^2 = \log h_N(\alpha) + O_{h,A}(k \log^2(q+2)).} \quad (29)$$

The small direction is unique in the critical layer. The exact covariance coisometry transforms (27) into

$$\boxed{M^{-1} = B_{\alpha,N} + (1 - \alpha)u \ell_N G_N^{-1} G_N(\alpha),} \quad (30)$$

where

$$B_{\alpha,N} = (1 - \alpha)B_N G_N^{-1} G_N(\alpha) + \alpha M^{-1} (G_N^J)^{-1} G_N(\alpha),$$

and

$$\|B_{\alpha,N}\| \leq \max \left(D_N, \frac{\sqrt{\mathbf{L}_k \mathbf{B}_k}}{R_0} \right) =: \bar{D}_N = e^{o_h(q)}.$$

All but the largest singular value of M^{-1} are at most $\bar{D}_N$.

If $L_N = \|M^{-1}\| > \bar{D}_N$, its top left singular line—the minimum right singular line of M —satisfies

$$\boxed{\|P_{\min,N} - P_{u,N}\|_1 \leq \frac{2\bar{D}_N}{L_N}.} \quad (31)$$

Indeed, projecting a unit top left singular vector orthogonally away from u leaves only $B_{\alpha,N}$ applied to its corresponding right vector.

The observation fraction is now protected at **every activation**:

$$\theta_N(\alpha) = \frac{\|\Lambda u\|_{Q_{B,N}(\alpha)}^2}{\|u\|_{G_N(\alpha)}^2} \geq \frac{L_{1,N}}{U_{1,N}},$$

so

$$-\log \theta_N(\alpha) = O_{h,A}(k \log^2(q + 2)). \quad (32)$$

For the measured full heat

$$\Theta_N(\tau; \alpha) = \text{Tr}_B \left[\Lambda e^{-\tau M^\dagger G_N(\alpha) M} L_{B,N}(\alpha) \right],$$

the complete finite estimate is

$$\boxed{\left| \frac{\Theta_N(\tau; \alpha)}{\theta_N(\alpha) e^{-\tau s_{\min,N}^2}} - 1 \right| \leq \frac{2\bar{D}_N s_{\min,N}}{\theta_N(\alpha)} + \frac{q-1}{\theta_N(\alpha)} e^{-\tau(\bar{D}_N^{-2} - s_{\min,N}^2)}. } \quad (33)$$

This is observation of the full heat, not heat of an independently compressed action—the distinction retained in the new proof.

Let

$$\kappa_{L/H}(\zeta) = \min(\beta_{L/H}, \zeta), \quad \tau_k(b) = e^{2q\ell_k + bq}.$$

Then

$$\tau_k(b) s_{\min,N}^2 = \exp\{q[b - \kappa_{L/H}(\zeta)] + o(q)\}.$$

Whenever

$$\kappa_H(\zeta) < b < \kappa_L(\zeta),$$

the two low heat factors tend to one and the high factors are doubly exponentially small. Thus

$$\boxed{\mathcal{R}\Theta_N(\tau_k(b); \alpha_k(\zeta)) > 0}$$

eventually, and

$$\boxed{\frac{1}{q} \log \mathcal{R}\Theta_N(\tau_k(b); \alpha_k(\zeta)) \longrightarrow 0.} \quad (34)$$

The unscaled limit is not assigned the value two, because the retained $\theta_N(\alpha)$ may decay.

For the much stronger activation $e^{-2\theta q}$, $\theta > 0$, equation (29) instead gives

$$s_{\min} = e^{-\theta q + o(q)},$$

while every other singular value is at least $e^{-o(q)}$. Thus the full heat at $e^{2\sigma q}$ tends to one for $0 < \sigma < \theta$, and to zero for $\sigma > \theta$. This identifies the soft inverse mode which persists even though the earlier, exponentially small-time heat gap has vanished.

6. An original observed arithmetic-action determinant is now evaluated

The observation of the constant class can vanish on an admitted period stratum. The correct way to avoid silently assuming otherwise is to use the **original first nonzero conductor moment**.

Let $v = \text{ord}_0 E_A \leq 80$. In the original centred coordinate,

$$C_A[y^v] = i^{-v} \mu_v \neq 0.$$

The phase and the binomial convention are those of the incoming conductor formula.

Define the original inverse-power subspace

$$W_r^{(v)} d = d_1[y^{-1}] + \sum_{j=2}^r d_j[y^{-(v+j)}].$$

Its exact arithmetic image is

$$\boxed{M^{v+1} W_r^{(v)} d = d_1[y^v] + \sum_{j=2}^r d_j[y^{-(j-1)}].} \quad (35)$$

The source is a coordinate subspace of U_{r+v} , so the incoming theorem makes $\Lambda W_r^{(v)}$ injective.

The target minimum has a uniform nonzero bound

Put $L = J(r-1)$, and assume $L+1 \leq q_-$. For any inverse-power coefficient vector,

$$C_A(y^v + U_{r-1}d) = i^{-v} \mu_v + \sum_{j,n} \frac{a_j d_n}{(y - \zeta_j)^n}.$$

Multiply by

$$B(y) = \prod_j (y - \zeta_j)^{r-1}.$$

The result is a degree- L polynomial with leading coefficient $i^{-v}\mu_v$.

Choose $L + 1$ original lower roots. Lagrange interpolation of the leading coefficient gives

$$\|C_A(y^v + U_{r-1}d)\|_2^2 \geq \frac{|\mu_v|^2}{(L+1) \max(1, (R_- + Z)/(2\delta))^{2L}}, \quad (36)$$

where

$$R_- = (k-8)R_0, \quad Z = \max_j |\zeta_j|.$$

Every lower-root difference is at least 2δ , and no pole coincides with a lower root. The conductor is a sum of weighted restrictions of the upper root-value vector, so

$$\|C_A w\|_2 \leq A_1 a_{\min}^{-k/2} \|w\|_{T_k}, \quad A_1 = \sum_j |a_j|.$$

Since $\ker \Lambda \subset \ker C_A$, this inequality applies to **every lift in the original observed fibre**, including every allowed kernel correction. Equations (6) and (36) therefore give

$$\boxed{d_- \leq \text{dist}_{Q_{B,N}(\alpha)}(\Lambda y^v, \Lambda U_{r-1})^2 \leq d_+,} \quad (37)$$

with

$$d_- = \frac{a_{\min}^k |\mu_v|^2}{\mathbf{L}_k A_1^2 (L+1) \max(1, (R_- + Z)/(2\delta))^{2L}},$$

$$d_+ = u_k \int_{\mathbb{R}} y^{2v} \sigma(y) dy.$$

Their logarithmic costs are

$$O_{h,A}(k \log^2 q + Jr \log q) = o(q).$$

This proves injectivity of the target in (35), without requiring $\Lambda_1 \neq 0$.

Define the fixed map between the two original observed subspaces:

$$\boxed{\mathcal{A}_r^{(v)}(\Lambda W_r^{(v)} d) = \Lambda M^{v+1} W_r^{(v)} d.} \quad (38)$$

At each cutoff, both subspaces use their restrictions of $Q_{B,N}(\alpha)$. The map is not $(\Lambda M L_{B,N})^{v+1}$.

Let $\mathcal{J}_{r,N}^{(v)}(\alpha)$ be its squared exterior-volume ratio. The inverse-power index sets

$$\{1, \dots, r-1\}, \quad \{1, v+2, \dots, v+r\}$$

have a common subset whose complementary dimensions sum to at most $2v+1$.

Factor their two Grams through that **same common subspace**. Its determinant cancels exactly. Every remaining attained quotient is controlled by (17). Then factor the target Gram through ΛU_{r-1} , retaining the low-column minimum (37). This proves

$$\boxed{\left| \log \mathcal{J}_{r,N}^{(v)}(\alpha) + \log h_N(\alpha) \right| \leq (2v+1) E_{r+v,N} + \max(|\log d_-|, |\log d_+|).} \quad (39)$$

The right side is $o(q)$, even for $r = k$. It is not r times an uncontrolled comparison of two large determinants.

Equation (39) proves (3). In particular, in the original canonical source,

$$\boxed{\mathcal{R} \log \mathcal{J}_{k,N}^{(v)}(0) = -\mathfrak{C}q + o_{h,A}(q) < 0} \quad (40)$$

eventually.

There is exactly one small singular direction in this map's critical layer. Its $r - 1$ inverse-power columns have source and target metrics controlled by the same h_N . The remaining Schur norm is between d_- and d_+ . Completing the square in this full Gram gives

$$-\log s_{\min}(\mathcal{A}_r^{(v)})^2 = \log h_N + o(q),$$

while its other singular values lie between $e^{-o(q)}$ and $e^{o(q)}$.

Therefore

$$\boxed{\mathcal{R} \operatorname{Tr}_{\Lambda W_r^{(v)}} e^{-\tau_k(b)(\mathcal{A}_r^{(v)})^\dagger \mathcal{A}_r^{(v)}} \longrightarrow 2} \quad (41)$$

on

$$\kappa_H(\zeta) < b < \kappa_L(\zeta).$$

Unlike (34), this is an unweighted limit: the full domain is already the specified observed inverse-power subspace.

7. A concrete simultaneous activation, arithmetic-volume, and heat calculation

Set

$$\beta_* = -2 - I_\square, \quad \zeta_\diamond = \beta_* + a_0,$$

$$\boxed{\alpha_k^\diamond = e^{-2q\ell_k - \zeta_\diamond q}, \quad \tau_k^* = e^{2q\ell_k + \beta_* q}.} \quad (42)$$

The heat time is exactly the incoming proof's original late time. Its endpoint inequalities imply

$$\beta_H < \beta_* < \zeta_\diamond < \beta_L.$$

No unknown Gram eigenvalue is used to choose either parameter.

For $r = k$, the three completed values are

$$\boxed{\mathcal{R} \log \det H_{k,N}^B(\alpha_k^\diamond) = [\mathfrak{C} - 2 \log(4/\pi)] kq + o_{h,A}(kq),} \quad (43)$$

$$\boxed{\mathcal{R} \log \mathcal{J}_{k,N}^{(v)}(\alpha_k^\diamond) = -[\mathfrak{C} - 2 \log(4/\pi)] q + o_{h,A}(q),} \quad (44)$$

and

$$\boxed{\mathcal{R} \operatorname{Tr} e^{-\tau_k^*(\mathcal{A}_k^{(v)})^\dagger \mathcal{A}_k^{(v)}} \longrightarrow 2.} \quad (45)$$

The same original observation, period, conductor moments, root polynomial, and cutoff windows occur in all three expressions.

8. The small-regularizer edge of the original spectral polynomial is also evaluated

Retain the preceding exact polynomial

$$\det(aI + M^{\dagger G_N(\alpha)} M) = \frac{\mathcal{P}_N(a, \alpha)}{\det(I + \alpha K_N)},$$

with

$$\mathcal{P}_N(0, \alpha) = |\det M|^2 \det(I + \alpha K_N).$$

These are its complete numerator and source-metric factors.

Since M is invertible,

$$\boxed{\frac{\mathcal{P}_N(a, \alpha)}{\mathcal{P}_N(0, \alpha)} = \det \left(I + a(M^{-1})^{\dagger} M^{-1} \right)}. \quad (46)$$

This is a determinant identity; the exchange between $TT^{\dagger}$ and $T^{\dagger}T$ does not assert an operator equality.

For

$$a_k(b) = e^{-2q\ell_k - bq},$$

the inverse rank-one decomposition gives the finite bound

$$0 \leq \log \det(I + a_k T^{\dagger} T) - \log(1 + a_k \|T\|^2) \leq (q-1)a_k \overline{D}_N^2, \quad T = M^{-1}.$$

Thus, including the threshold,

$$\boxed{\frac{1}{q} \log \frac{\mathcal{P}_N(a_k(b), \alpha_k(\zeta))}{\mathcal{P}_N(0, \alpha_k(\zeta))} \longrightarrow [\min(\beta_{L/H}, \zeta) - b]_+}. \quad (47)$$

At $a_k = 1/\tau_k^*$ and $\alpha_k = \alpha_k^{\diamond}$, the exact two-metric spectral identity now gives

$$\boxed{\mathcal{R} \left[\log \frac{\mathcal{P}_N(a_k, \alpha_k)}{\mathcal{P}_N(a_k, 0)} - \Delta_N(\alpha_k) \right] = -2 \log(4/\pi) q + o_{h,A}(q)}. \quad (48)$$

The full source-metric cost Δ_N remains in the receiver. This calculation evaluates the **small-regularizer** boundary, complementary to the earlier large-regularizer/large-singular-value calculation.

The new input therefore has been carried through the source minimum, the observation, the inverse arithmetic singular direction, a specified growing-rank observed arithmetic action, and the complete spectral-polynomial identity. The finite comparison is (17); its critical coefficient is (2); the action-volume and heat values are (43)–(45).

These results do not assign the separate first-order projected-current phase or the independent proper-source Z_R minimum. In particular, the negative logarithmic action-volume return in (44) is not by itself an RH contradiction: it is the calculated return of the explicit map (38), with its original metrics and value spaces. The shared page was not retrievable in the preceding attempt; the continuation above uses the complete attached proof and does not depend on unseen content from that page.